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Political Discourse on and Economic Impact of Migration: An Empirical Analysis

Masterthesis

presented by

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List of Abbreviations

NLP	natural language processing
PCA	Principal Component Analysis
XML	Extensible Markup Language
TF-IDF	Term Frequency-Inverse Document Frequency
PV-DBOW	Paragraph Vector - Distributed Bag of Words
LLM	large language models
OECD	Organisation for Economic Co-operation and Development

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1 Introduction

“We have to stop the immigrant invasion!” Such rhetoric, deliberately provocative as it may be, has become increasingly familiar in contemporary political debate. Over the past two decades, immigration has gained growing prominence in political and policy debates across several advanced democracies (Ivarsflaten 2008). Its salience stems not only from the direct economic and social consequences of migration but also from its capacity to serve as a focal point through which broader concerns about employment, demographic change, public services, and social cohesion are articulated. In particular, immigration has become an important issue through which right-wing populist parties mobilize broader economic and social grievances (Ivarsflaten 2008; Fang and Speer n.d.).

Political actors, however, frequently invoke similar underlying economic conditions to justify fundamentally different positions on immigration. In South Africa, for example, the anti-immigrant movement Operation Dudula has accused immigrants of taking jobs from citizens against the backdrop of persistently high unemployment (Yu 2025; Hlatshwayo 2023). In France, the Reconquête party has employed the “great replacement” narrative, linking immigration to declining birth rates and framing demographic change as a threat (Ekman 2022). In Germany, the AfD has adopted predominantly restrictive positions on immigration, pointing to the neglect of domestic education, vocational training, and technological automation in favor of relying on a lower-cost foreign workforce (Deutscher Bundestag 2023). In the United Kingdom, Reform UK has linked immigration to pressures in the housing market and advocated restrictions on access to social housing for foreign nationals (Reform UK 2025). Conversely, political actors supporting immigration frequently emphasize its potential economic benefits, including alleviating labour shortages, contributing to economic output, and mitigating the consequences of population ageing (Ekman 2022). Thus, the same economic conditions—including unemployment, demographic ageing, labour shortages, and housing pressures—can be interpreted and mobilized to support opposing political narratives.

This tension raises a question that extends beyond the established relationship between economic conditions and political discourse: whether the relationship may also operate in the opposite direction. Existing research documents substantial macroe-

conomic effects of migration, including effects on employment, wages, consumption, investment, and output (Maffei-Faccioli and Vella 2021; Ortega and Peri 2009; Abramitzky and Boustan 2017). A large literature likewise examines how economic conditions shape public and political attitudes toward immigration (Abdelatif and Andriantomanga 2026; Scheve and Slaughter 2001; Hainmueller and Hopkins 2014; Dustmann and Preston 2007). Comparatively less attention, however, has been devoted to whether immigration-related political discourse itself contains information about subsequent migration and macroeconomic dynamics. Such a relationship is plausible because political discourse may contribute to shaping expectations about future policy, economic conditions, and migration before these expectations are reflected in realized outcomes. At the same time, economic and migration developments may feed back into political discourse, creating a dynamic relationship in which neither direction can be considered in isolation.

This thesis investigates whether immigration-related political sentiment and macroeconomic conditions are dynamically related, allowing for feedback in both directions rather than imposing a one-directional relationship *a priori*. To this end, we extend the Bayesian Vector Autoregression (BVAR) framework of (Maffei-Faccioli and Vella 2021), which examines the macroeconomic effects of migration shocks in Germany, by incorporating a novel text-based measure of immigration sentiment derived from speeches in the German Bundestag. The sentiment measure is constructed using a purpose-built dictionary designed to distinguish between restrictive and welcoming positions toward immigration and is subsequently incorporated into the VAR system as an endogenous variable alongside migration and macroeconomic indicators. This framework makes it possible to examine whether changes in immigration-related political sentiment contain predictive information about subsequent developments in migration and macroeconomic conditions, while simultaneously accounting for the possibility that economic and migration developments influence subsequent political sentiment.

The main methodological contribution of this thesis is therefore the integration of parliamentary text data into a multivariate macroeconomic time-series framework. Rather than treating political discourse solely as an outcome of economic and migration conditions, the analysis treats sentiment expressed in parliamentary debates as a measurable time-varying variable that may contain information about future developments. This approach combines quantitative text analysis with Bayesian time-series

modelling and provides a framework for studying the temporal interaction between political discourse, migration, and macroeconomic conditions. Importantly, the empirical framework does not presume that political sentiment causes subsequent economic or migration outcomes. Instead, it allows the temporal structure of the data to reveal whether systematic dynamic relationships and predictive interactions exist between these variables.

Germany provides a particularly suitable setting for this analysis. First, the study can build directly on the empirical framework developed by (Maffei-Faccioli and Vella 2021), allowing the analysis to extend an established examination of migration and macroeconomic dynamics. Second, Germany has experienced substantial variation in both migration flows and the political salience of immigration. The 2015–2016 migration episode represented a major turning point in German immigration politics and generated extensive parliamentary debate. More recently, the large-scale displacement following Russia’s invasion of Ukraine in 2022 produced another substantial increase in migration and renewed political attention to immigration. These episodes, together with longer-term changes in migration and economic conditions, provide substantial temporal variation for examining whether shifts in immigration-related political sentiment coincide with, precede, or follow changes in migration and macroeconomic indicators.

The remainder of this thesis is organized as follows. Section 2 reviews the relevant literature on migration, macroeconomic dynamics, political attitudes, and political discourse. Section 3 presents the parliamentary text data and describes the data preparation procedure, the construction and refinement of the immigration sentiment dictionary, the construction of the sentiment measure from Bundestag speeches, and its validation. Section 4 introduces the Bayesian VAR framework, presents the macroeconomic and migration variables used in the analysis, discusses the identification strategy, and presents and discusses the empirical results together with a series of robustness checks. Section 5 concludes.

2 Literature Review

Migration has long been an important economic and social force. States have used migration policy to address labour shortages, demographic change, and broader eco-

nomic and territorial goals (Hatton and Williamson 2005; Fellini et al. 2023; Emmer 2012; Okia 2023; Qvarnström 2014; Adamson and Tsourapas 2020; Adamson and Thiollet 2025). After the Second World War, Western European countries increasingly used organised labour recruitment, including the “guest worker” programmes involving Southern Europe and the Maghreb (Fellini et al. 2023). Today, migration policy is closely linked to labour demand, demographic change, security, integration, and migrants’ rights (Hollifield 2004).

Migration decisions are often explained by the push–pull model. Unemployment, poverty, conflict, and political instability can push people to leave, while better economic and social conditions can attract migrants to destination countries (Lee 1966; Parliament 2020). Information and transport can further reduce the costs of migration (Dekker, Engbersen, and Faber 2016; Pesando et al. 2021; Fellini et al. 2023). At the same time, ageing populations and labour shortages create demand for foreign workers in advanced economies (OECD 2024b; OECD 2024a). Migration can nevertheless generate concerns about public services, housing, security, cultural identity, and economic pressure (Huysmans 2000; Young, Loebach, and Korinek 2018; OECD 2024a). This tension provides an important context for political discourse on immigration.

Germany provides a useful case for studying these dynamics. The Gastarbeiter agreements of the 1950s and 1960s framed immigration mainly as a response to labour shortages. In the early 1990s, rising asylum applications coincided with more restrictive rhetoric and the 1993 asylum compromise (Borkert and Bosswick 2007). Around 2000, policy became more open, including birthright citizenship and the Süßmuth Commission (Borkert and Bosswick 2007). The 2015–2016 refugee inflow again transformed political debate, while the arrival of refugees from Ukraine in 2022 generated substantial institutional and public support (Trauner and Turton 2017; Brücker 2022). These episodes illustrate that German immigration discourse can change substantially with political and social conditions.

Political actors also differ in how they frame immigration. Positive and negative views coexist and respond to political and social events (Card et al. 2022). Threat-based messages can increase concerns about economic competition, security, and national or cultural identity (Brader, Valentino, and Suhay 2008; Esses, Jackson, and Armstrong 1998). Parliamentary discourse can therefore reveal political preferences while also reflecting broader economic and social concerns. This creates an important

empirical question: whether political discourse merely responds to changing conditions or also contains information about subsequent developments.

The growth of digitised parliamentary records makes such questions measurable at scale (Grimmer and Stewart 2013). Dictionary methods are transparent and efficient, but can miss context, negation, and reported speech Kostikova et al. (2024). Supervised models and large language models can capture more context, but require more data, computation, and often manual labelling. Kostikova et al. (2024), for example, use large language models to measure solidarity and anti-solidarity toward migrants in Reichstag and Bundestag speeches over 155 years.

This thesis therefore combines the transparency of a dictionary approach with corpus-specific semantic expansion. The dictionary is manually seeded and refined using word embeddings, allowing the measure to adapt to the language of the corpus while retaining an interpretable construction. As a robustness check, immigration sentiment is also estimated with GPT-4.1-nano and GPT-5-nano following (Kostikova et al. 2024). Related work on UK parliamentary discourse likewise finds shifts toward more securitised and border-control framing and less emphasis on integration (Ghafouri et al. 2025).

A related literature demonstrates that political text can be transformed into quantitative indicators with economic information. Latifi et al. (2024) construct a fiscal-policy sentiment measure from Bundestag speeches and incorporate it into a recursively identified VAR. They find that changes in fiscal sentiment are associated with subsequent changes in government spending and other macroeconomic variables. Tillmann (2025) further shows that monetary tightening affects fiscal sentiment differently for government and opposition speakers. Together, these studies suggest that parliamentary text can provide economically relevant information and that political position may matter for its interpretation.

The closest study is Abdelatif and Andriantomanga (2026), who construct a migration sentiment index from German Bundestag speeches for 1991–2018. Using openness and restriction terms around migration anchors and correcting for negation, they find that higher economic and migration-related uncertainty is followed by more restrictive parliamentary rhetoric. Their study therefore provides evidence for one direction of the relationship: economic and migration conditions shape parliamentary sentiment. This thesis examines the complementary direction by asking whether par-

liamentary immigration sentiment contains information about subsequent economic and migration dynamics.

The migration literature provides the other side of this relationship. Franz, Oser, and Winker (1994) find that the effects of migration depend on economic conditions and market regimes. Using a structural VAR, Maffei-Faccioli and Vella (2021) find that net migration shocks affect vacancies, wages, house prices, consumption, investment, net exports, output, and unemployment. These findings establish that migration itself can have broad economic effects, while also suggesting that its consequences depend on the wider economic environment.

Taken together, the literature establishes three relevant relationships: migration can affect economic outcomes, economic and migration conditions can shape immigration attitudes and parliamentary rhetoric, and political text can contain measurable economic information. What remains less explored is whether immigration sentiment expressed in parliamentary debates contains information about subsequent migration and macroeconomic dynamics. In particular, it is unclear whether sentiment provides information about future developments or primarily reflects conditions that are already underway.

This thesis addresses this gap by constructing immigration-sentiment measures from German Bundestag speeches and incorporating them into a Bayesian vector autoregressive framework based on Maffei-Faccioli and Vella (2021). The analysis distinguishes government and opposition speakers and examines robustness across sample periods, recursive orderings, and alternative sentiment measures. It therefore connects migration economics with computational political text analysis and investigates whether parliamentary immigration discourse is systematically related to subsequent economic and migration dynamics.

3 Text data

3.1 Data Sources and Corpus Construction

We use a compiled dataset of speeches from the German Bundestag, provided by Department 02: Business Studies and Economics at Justus-Liebig-Universität Gießen. The dataset comes from parliamentary speeches available as digital Extensible Markup

Language (XML) files. These raw files are not ready for natural language processing (NLP) analysis and need extraction and preprocessing first.

Our corpus combines two sub-datasets. Both end up with the same structure, but they were extracted differently, since the Bundestag changed its XML format over time. Table 1 shows both sub-datasets.

Table 1: Overview of the two sub-datasets underlying the combined corpus.

	Sub-dataset 1	Sub-dataset 2
Coverage	Q1 1960 – Oct. 2017	Oct. 2017 – Q2 2025
XML structure	Unstructured (continuous debate text)	Structured (individual speeches)
Extraction tool	Custom workflow, Latifi et al. (2024)	pybundestag package

The first sub-dataset covers the first quarter of 1960 to October 2017. In this period, the XML files store whole debates as continuous text, with only limited metadata. To turn these into speech-level data, Latifi et al. (2024) built a custom extraction workflow, first designed for proceedings from 7 September 1949 to 7 September 2021. Each extracted speech is linked to its session and date, and includes the speaker’s role and party.

The second sub-dataset covers October 2017 to the second quarter of 2025. After a format change in the 19th legislative period, records became structured, so speeches and their metadata could be extracted directly. For this period, Latifi et al. (2024) used the `pybundestag` package. The result matches the structure of the first sub-dataset. Combining both gives one consistent speech-level corpus from Q1 1960 to Q2 2025.

Preprocessing was done by the data provider, following Latifi et al. (2024); we received the corpus already cleaned. The steps were: (i) removing irrelevant speeches and speeches shorter than 100 or longer than 3,573 tokens; (ii) lemmatization; (iii) normalizing German umlauts and the Eszett (ß); (iv) removing line breaks, digits, and special characters; (v) lowercasing all text; (vi) removing single-character tokens and tokens longer than 30 characters; and (vii) removing stop words.¹

This pipeline produces a final dataset of exactly 260,660 parliamentary speeches, covering Q1 1960 to Q2 2025.

¹Stop words are very common words that carry little meaning and add little to text analysis.

3.2 Migration Speech Identification and Dictionary Construction

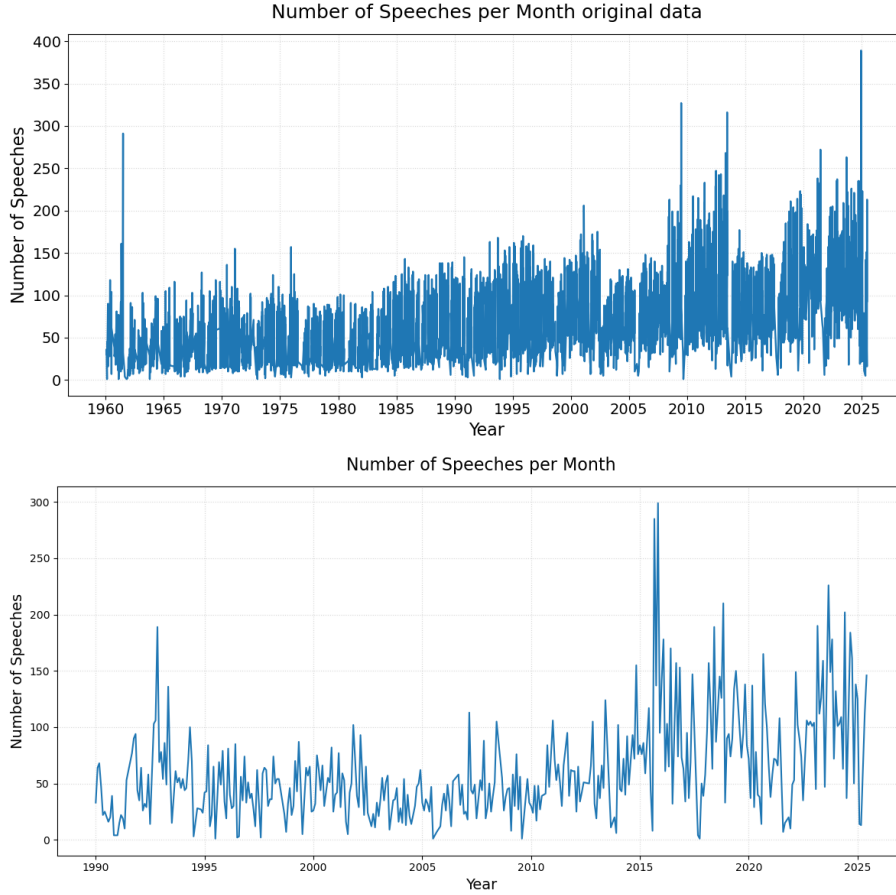


Figure 1: Monthly number of parliamentary speeches (top: original data; bottom: immigration-related speeches after keyword filtering).

To make sentiment scores easier to interpret, and to reduce computing time, we keep only speeches that clearly discuss immigration. We identify these using keyword filtering, based on the keyword list from Kostikova et al. (2024). Since our filter uses Python’s `re` module for pattern matching, we merge word forms that share the same root (e.g., *Flüchtling*, *Flüchtlinge*, *Flüchtlingen*) into a single stem, so no two entries overlap. This reduces the original list of 32 word forms (Table 2) to 13 stems covering the same meaning without duplication (Table 3).

This filtering reduces the corpus from 260,660 to 28,662 speeches, about 11% of all speeches. Figure 1 shows the counts before and after filtering. While total speeches have grown steadily over time, immigration-related speeches have grown even more, especially during periods of high migration.

Table 2: Original keyword list (32 inflected word forms), following Kostikova et al. (2024).

Keywords			
Flüchtlinge	Ausländer	Flüchtlingen	Zuwanderung
Vertriebenen	Ausländern	Asylbewerber	Migranten
Migration	Heimatvertriebenen	Aussiedler	Einwanderung
Ansiedler	Vertriebene	Zuwanderer	Asylbewerbern
Flüchtling	Heimatvertriebene	Sowjetzonenflüchtlinge	Aussiedlern
Einwanderer	Asylsuchenden	Asylsuchende	Bürgerkriegsflüchtlinge
Zuwanderern	Ansiedlern	Migrantinnen	Vertriebener
Emigranten	Kriegsflüchtlinge	Ausländerinnen	Immigranten

Table 3: Reduced list of 13 word stems used for regular-expression matching. Word forms sharing a root were merged into one stem, so no entry overlaps with another.

Word stems			
migration	zuwander	einwander	asyl
fluechtling	gefluechtet	schutzsuchend	auslaend
vertrieben	migrant	aussiedler	ansiedler
buergerkriegs			

From this reduced corpus, we build two sentiment dictionaries: a welcoming (open) dictionary and a restrictive dictionary. We build them in three steps: manual seed selection, Word2Vec-based expansion, and manual cleaning.

Step 1: Manual seed selection. We manually pick speeches that reflect different immigration positions, based on the documented history of German parties on this issue: 200 AfD speeches from 2017–2025, 94 CDU/CSU speeches from 1990–1994, and 500 CDU/CSU, SPD, and FDP speeches from the 2015–2018 refugee period. Close reading of these speeches gives us 655 seed words for the welcoming dictionary and 872 for the restrictive dictionary.

Step 2: Word2Vec-based expansion. We train a skip-gram Word2Vec model on the immigration-related corpus (vector size 100, context window 7, minimum word count 10, 50 training epochs, fixed random seed). We expand the dictionaries iteratively: in each round, we use a small, topic-focused seed list (for example, terms about the economic side of migration policy). For each seed word in the model’s vocabulary, we find its 20 nearest neighbors by cosine similarity. We keep a candidate word only if (i) its similarity to the seed is at least 0.72, (ii) it appears in a Term Frequency-

Inverse Document Frequency (TF-IDF) vocabulary built on the immigration corpus ($max_features = 80,000$, $min_df = 2$), and (iii) it is a purely alphabetic token longer than three characters. Words that appear as neighbors of seeds from both dictionaries are treated as ambiguous and dropped from both. We merge the kept words from each round into the growing dictionaries, ending with 755 welcoming words and 968 restrictive words.

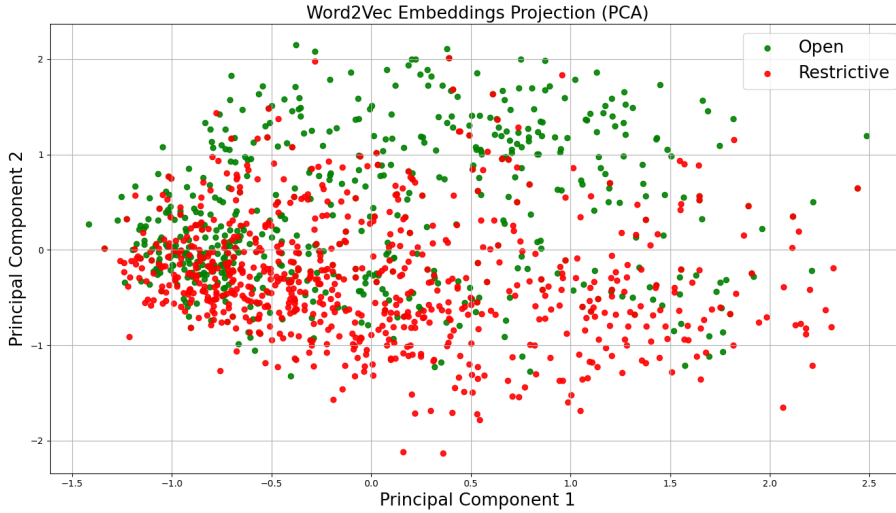


Figure 2: PCA projection of the expanded immigration dictionaries before cleaning (restrictive = red, welcoming = green).

To sharpen the split between the two dictionaries, we project the word embeddings onto two dimensions using Principal Component Analysis (PCA), which keeps most of the variance while giving an interpretable view of the embedding space. As Figure 2 shows, restrictive terms (red) mostly sit lower in the projection, while welcoming terms (green) sit higher. The interval $[-0.48, 0.5]$ on the y-axis holds most ambiguous words. We keep restrictive words with a y-coordinate below -0.48 and welcoming words above 0.5 , dropping everything in between. This gives cleaner dictionaries (Figure 3) of 288 restrictive and 241 welcoming terms.

Step 3: Manual cleaning. We then manually clean the dictionaries, prompted by an early check of average sentiment scores by party (Section 3.3). This check found implausibly negative scores for parties known to hold welcoming positions, especially Bündnis 90/Die Grünen and Die Linke. For each of these parties, we randomly sampled speeches with a sentiment score below zero and read them individually.

This revealed two problems. First, restrictive words were sometimes used criti-

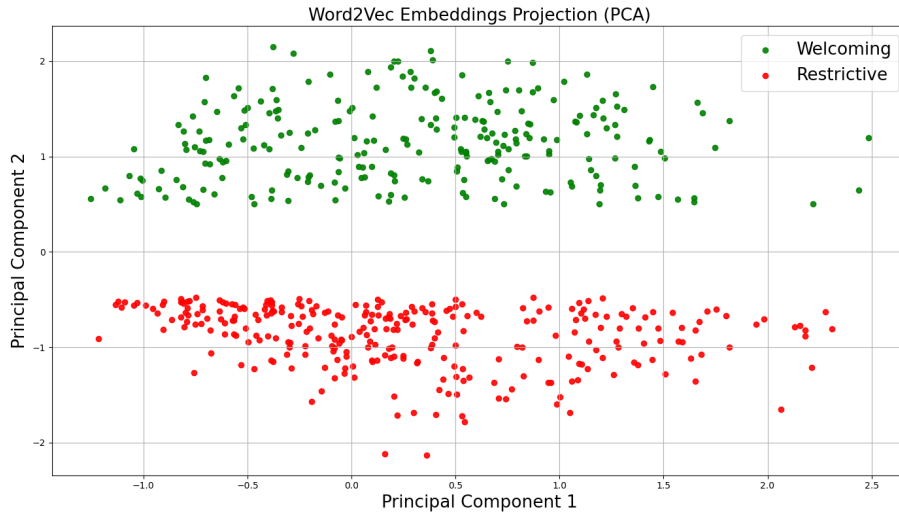


Figure 3: PCA projection of the refined dictionaries after y-coordinate threshold cleaning (restrictive = red, welcoming = green), showing clearer separation between the two stances.

cally, for example when a speaker quoted or condemned anti-immigration rhetoric rather than expressing it themselves — a nuance our embedding-based measure cannot capture. Second, some dictionary words carried purely administrative or legal meaning (e.g., *Familiennachzug*, *Asylrecht*, *Ausländer*) or no clear stance at all (e.g., *abschieben*, *Abschiebung*, *Zurückweisung*, *Migrationswelle*, *Migrationsstrom*), adding noise unrelated to the speaker’s actual position. We removed all such entries, retrained the model, and recomputed the scores.

This cleaning was not done to force parties into a preset ranking: after this step, Bündnis 90/Die Grünen’s average sentiment became less negative, while Die Linke’s stayed negative (Figure 4). This shows the correction came from real misclassifications found by close reading, not from adjusting until a target ranking appeared. We revisit this in the robustness check (Section 4.5).

This gives a final welcoming dictionary of 222 words, covering skilled labor, the economy, the labor market, human rights, humanitarian aid, protection, integration, education, language learning, citizenship, social cohesion, and diversity and inclusion. The final restrictive dictionary has 178 words, covering border security, state control, repatriation, deportation, legal enforcement, capacity limits, labor market protection, and concerns about social integration ².

²To conserve space in the main document, the complete lists of welcoming and restrictive dictionary terms are provided in the replication code and supplementary data folder accompanying this thesis.

Table 4: Evolution of the welcoming and restrictive vocabularies

Stage	Welcoming	Restrictive	Total
Initial seeds	655	872	1527
Word2Vec expansion	755	969	1724
PCA refinement	241	288	529
Final vocabulary	222	178	400

3.3 Sentiment Index Construction

We build an immigration sentiment index from the reduced corpus. To capture semantic similarity in natural language, we need a method that turns sentences, paragraphs, and whole documents into numeric vectors while keeping their meaning. Since our corpus has thousands of speeches, Doc2Vec is a good fit: it is an unsupervised, neural-network method that learns vector representations of documents in a high-dimensional space (Le and Mikolov 2014).

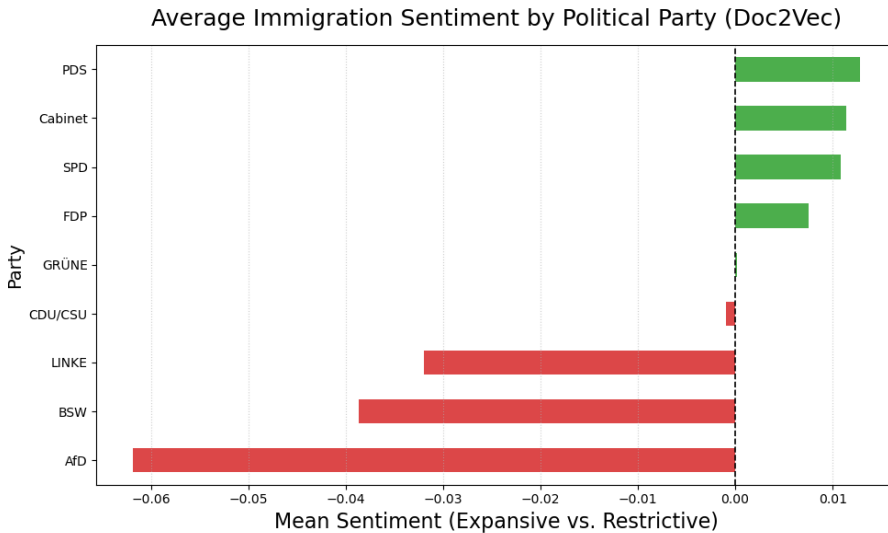


Figure 4: Average restrictive immigration sentiment by political party (negative values indicate frequent use of restrictive anti-immigration vocabulary, whether framed defensively or critically).

Instead of simply counting dictionary words, we embed both speeches and dictionary terms in one shared vector space, and define each speech’s stance as its cosine similarity to two reference vectors — one per dictionary. We use a rolling-window Doc2Vec procedure to avoid look-ahead bias: speeches in a given quarter are always scored using a model trained only on earlier data.

For each quarter t , we train a Paragraph Vector - Distributed Bag of Words (PV-DBOW)

Doc2Vec model ($\text{dm}=\mathbf{0}$, with word vectors trained jointly via $\text{dbow_words}=1$), vector size 100, context window 5, minimum word count 1, initial learning rate 0.025, and 50 training epochs, using only speeches from the preceding 40 quarters (ten years), $[t - 40, t)$. This follows the rolling-window design of Latifi et al. (2024) for fiscal sentiment, adapted here for immigration. Figure 5 shows the procedure.

Using this model, we build two reference vectors — one for each dictionary — by averaging the embeddings of all dictionary words present in that window’s vocabulary (for multi-word entries, we sum the word embeddings). We then infer document vectors for all speeches in the following, held-out quarter t , and compute their cosine similarity to each reference vector, giving a $\text{similarity}_{\text{welcoming}}$ and a $\text{similarity}_{\text{restrictive}}$ score per speech.

We start the rolling training from Q1 1980 onward, since immigration-related discourse becomes more consistent from this point. With the ten-year window requirement, this gives out-of-sample sentiment scores from Q1 1990 to Q2 2025.

The sentiment score of speech i is the difference between its two similarities:

$$\text{sentiment}_i = \text{similarity}_{\text{welcoming},i} - \text{similarity}_{\text{restrictive},i}. \quad (3.1)$$

A positive score means a speech is closer to the welcoming reference vector; a negative score means the opposite.

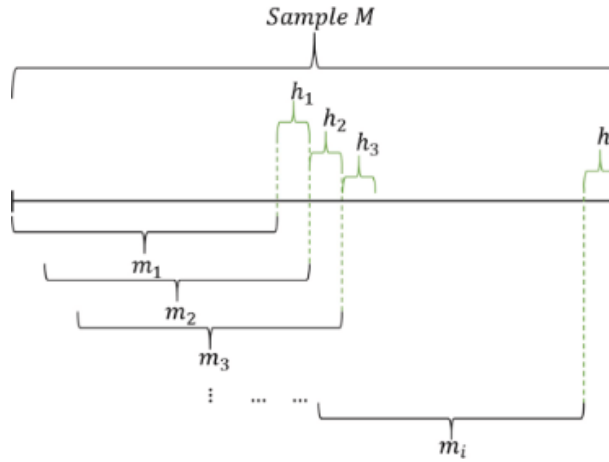


Figure 5: Rolling-window estimation for the Doc2Vec model. The corpus is split into fixed training windows (m) and following inference windows (h), shifted forward by one period each time. Adapted from Latifi et al. (2024).

3.4 Validation of the Sentiment Measure

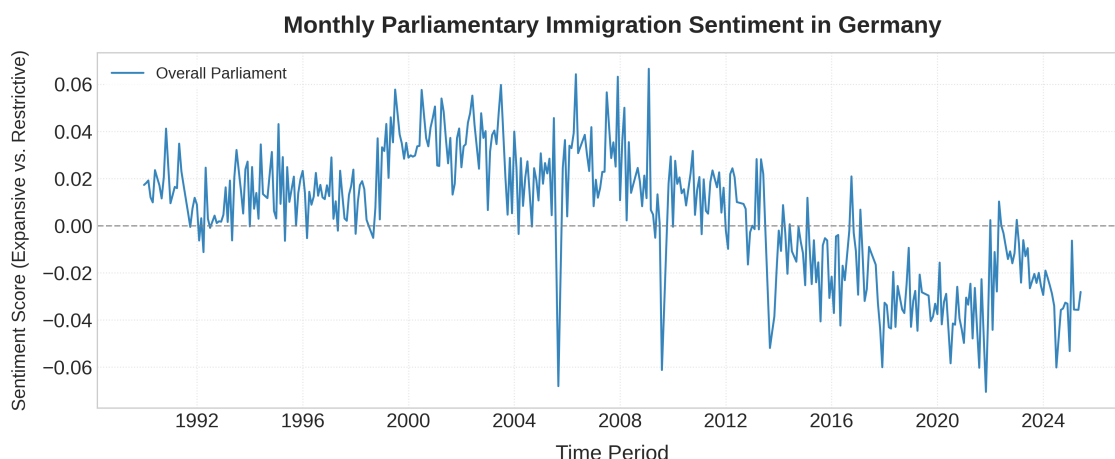


Figure 6: Monthly trend of overall parliamentary immigration sentiment in Germany. Positive scores show a net-expansive vocabulary; negative scores show more restrictive terminology (whether defensive or critical).

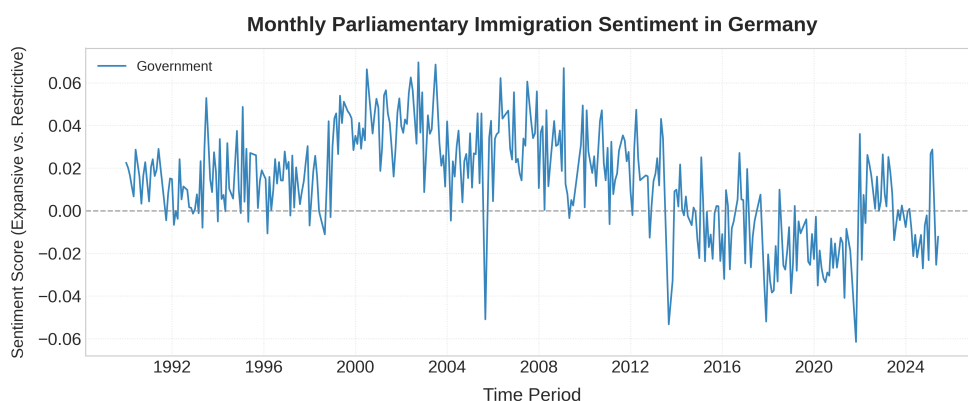


Figure 7: Monthly trend of government parliamentary immigration sentiment in Germany.

We check the plausibility of our sentiment measure by comparing it to well-known shifts in German immigration politics. A close match between the estimated series and known historical events is qualitative evidence that the measure captures real variation in political discourse, not just noise.

Overall sentiment shows two broad phases: mostly positive from 1990 to 2014, then clearly negative from 2014 onward (Figure 6). Government and opposition sentiment move together for most of the period, diverging mainly around 2006–2007 and 2018–2022 (Figure 9).

A clear dip appears between 1992 and 1994, matching the more restrictive rhetoric of that time aimed at reducing asylum applications (Borkert and Bosswick 2007). Sentiment turns more positive around 2000, matching the introduction of birthright citizenship (*jus soli*) and the Süßmuth Commission’s work (Borkert and Bosswick 2007). Between 2014 and 2021, government sentiment swings more sharply (Figure 7), while opposition sentiment trends more negative (Figure 8) — patterns credibly linked to the Syrian refugee crisis (Trauner and Turton 2017), the COVID-19 pandemic, and the 2021 federal election, where immigration was a major campaign topic and public debate turned more restrictive. The positive peak in overall sentiment in 2022 (Figure 6) realistically reflect solidarity with Ukrainian refugees (Brücker 2022).

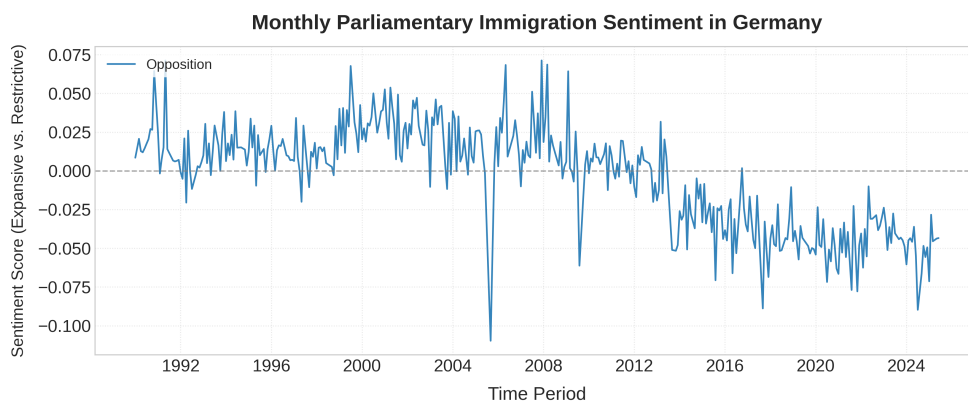


Figure 8: Monthly trend of opposition parliamentary immigration sentiment in Germany.

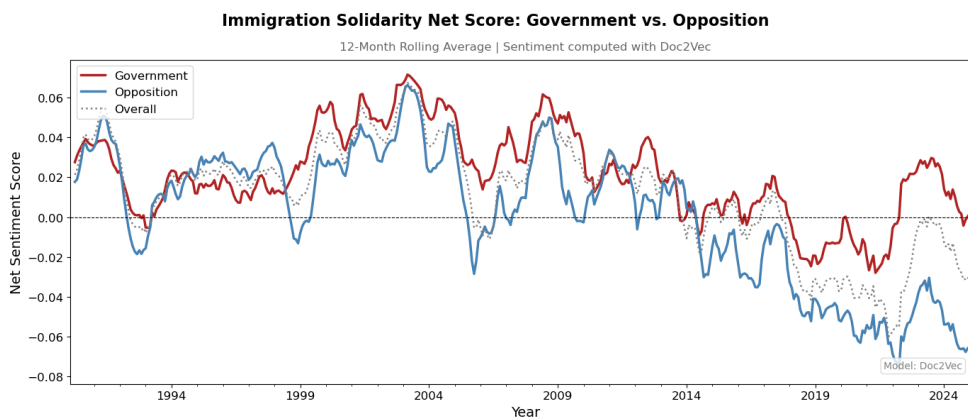


Figure 9: Monthly trend of immigration sentiment for the overall parliament, government, and opposition in Germany.

Party-level results show large differences in immigration-related discourse. The AfD shows mostly negative sentiment throughout its observed period, while parties tradi-

tionally seen as pro-immigration show higher average scores, though variation within each party remains substantial (Appendix 36).

4 Estimating the macroeconomic effects

We now investigate the macroeconomic effects of immigration sentiment as reflected in the textual data. For that purpose, we augment the baseline VAR model from the study (Maffei-Faccioli and Vella 2021) by our new sentiment series.

4.1 VAR Model

This study employs a Bayesian vector autoregressive (VAR) framework to analyse the dynamic relationships among the selected macroeconomic and political variables. The model is estimated over different sample periods in order to examine the robustness of the results and to distinguish between the pre-pandemic and extended sample periods. Specifically, the analysis considers the periods 2006–2019, 2007–2019, and 2007–2025.

For the pre-pandemic samples, namely 2006–2019 and 2007–2019, we estimate a standard reduced-form VAR model with p lags and n endogenous variables:

$$\mathbf{y}_t = \boldsymbol{\mu} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t, \quad (4.1)$$

where $\mathbf{y}_t$ denotes an $n \times 1$ vector of endogenous variables, $\boldsymbol{\mu}$ is an $n \times 1$ vector of intercept terms, and $\mathbf{A}_i$, for $i = 1, \dots, p$, are $n \times n$ autoregressive coefficient matrices. The reduced-form disturbance vector $\mathbf{u}_t$ is assumed to follow a multivariate normal distribution,

$$\mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \quad (4.2)$$

where $\boldsymbol{\Sigma}$ is a positive-definite $n \times n$ variance–covariance matrix satisfying

$$E(\mathbf{u}_t \mathbf{u}_t') = \boldsymbol{\Sigma}. \quad (4.3)$$

The reduced-form disturbances are assumed to be serially uncorrelated at all non-zero leads and lags.

For the extended sample covering the period 2007–2025, the baseline VAR specification is augmented by an exogenous COVID-19 dummy variable in order to account for the exceptional economic and social disturbances associated with the pandemic. The extended model is therefore specified as

$$\mathbf{y}_t = \boldsymbol{\mu} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{B} \mathbf{x}_t + \mathbf{u}_t, \quad (4.4)$$

where $\mathbf{x}_t$ denotes the vector of exogenous variables and $\mathbf{B}$ contains the corresponding coefficients. In the present application, $\mathbf{x}_t$ consists of a single COVID-19 dummy variable, such that

$$COVID_t = \begin{cases} 1, & \text{for observations between March 2020 and October 2021,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.5)$$

The COVID-19 dummy is included only in the estimation based on the 2007–2025 sample. It is treated as an exogenous control variable and does not form part of the endogenous VAR system. Consequently, it does not enter the autoregressive dynamics represented by the coefficient matrices $\mathbf{A}_1, \dots, \mathbf{A}_p$. The VAR model can be written in compact matrix form as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\Pi} + \mathbf{U}, \quad (4.6)$$

where

$$\mathbf{Y} = (\mathbf{y}_{p+1}, \dots, \mathbf{y}_T)' \quad (4.7)$$

is a $(T - p) \times n$ matrix containing the observations of the endogenous variables.

For the baseline VAR specification, the regressor matrix is given by

$$\mathbf{X} = (\mathbf{1}, \mathbf{Y}_{-1}, \dots, \mathbf{Y}_{-p}), \quad (4.8)$$

where $\mathbf{1}$ is a $(T - p) \times 1$ vector of ones and

$$\mathbf{Y}_{-j} = (\mathbf{y}_{p+1-j}, \dots, \mathbf{y}_{T-j})' \quad (4.9)$$

is a $(T - p) \times n$ matrix containing the j th lag of the endogenous variables. Thus, in the baseline specification, $\mathbf{X}$ has dimension

$$(T - p) \times (1 + np). \quad (4.10)$$

For the extended 2007–2025 specification, the exogenous COVID-19 dummy is in-

cluded in the regressor matrix:

$$\mathbf{X} = (\mathbf{1}, \mathbf{X}_{COVID}, \mathbf{Y}_{-1}, \dots, \mathbf{Y}_{-p}). \quad (4.11)$$

Since only one exogenous variable is included, the corresponding regressor matrix has dimension

$$(T - p) \times (2 + np). \quad (4.12)$$

More generally, with k exogenous variables, the dimension of the regressor matrix is $(T - p) \times (1 + k + np)$.

The coefficient matrix for the baseline specification is defined as

$$\mathbf{\Pi} = (\boldsymbol{\mu}, \mathbf{A}_1, \dots, \mathbf{A}_p)', \quad (4.13)$$

whereas, for the extended specification including the COVID-19 dummy, it is given by

$$\mathbf{\Pi} = (\boldsymbol{\mu}, \mathbf{B}, \mathbf{A}_1, \dots, \mathbf{A}_p)'. \quad (4.14)$$

The error matrix is defined as

$$\mathbf{U} = (\mathbf{u}_{p+1}, \dots, \mathbf{u}_T)', \quad (4.15)$$

and has dimension $(T - p) \times n$. We estimate the VAR model using the variables in levels. The KPSS results indicate that most variables are non-stationary in levels (see Tables 7 and 9), while the Johansen trace test provides evidence of long-run relationships among the variables (see Tables 8 and 10 in the Appendix). Given the objective of preserving these long-run relationships and following the level-VAR approach adopted in the reference study, the VAR models are estimated in levels. The resulting impulse responses are therefore interpreted primarily as dynamic associations rather than as estimates of a structural causal mechanism (Sims, Stock, and Watson 1990).

4.1.1 Bayesian Estimation

Both the baseline VAR models and the extended specification including the COVID-19 dummy are estimated within a Bayesian framework.

Lag length was assessed using the Akaike Information Criterion (AIC), Hannan–

Quinn Information Criterion (HQ), Schwarz Information Criterion (SC), and Final Prediction Error (FPE). As shown in Table 5, the criteria disagree substantially: AIC and FPE favor long lag structures, particularly in the 2007–2019 and 2007–2014 sub-samples, while SC consistently favors one or two lags across all samples. We follow the Schwarz Criterion throughout, both for its parsimony and to guard against over-fitting given the limited sample size. This yields one monthly lag for the original 2006–2019 specification, two lags for the reconstructed 2007–2025 sample, and one lag for the reconstructed 2007–2019 and 2007–2014 sub-samples.

Table 5: VAR Lag Length Selection

Sample	AIC	HQ	SC	FPE	Final Lag (SC)
Original study (2006–2019)	3	2	1	3	1
Reconstructed, 2007–2025	3	3	2	3	2
Reconstructed, 2007–2019	10	2	1	9	1
Reconstructed, 2007–2014	5	2	1	4	1

Notes: AIC denotes the Akaike Information Criterion, HQ the Hannan–Quinn Information Criterion, SC the Schwarz Information Criterion, and FPE the Final Prediction Error. Reported values indicate the lag order minimizing each criterion. The lag order selected by SC is retained in the final specification for each sample.

The objective is to generate posterior draws (one thousand) of the coefficient matrix $\mathbf{\Pi}$ and the covariance matrix $\mathbf{\Sigma}$ from their joint posterior distribution. According to Bayes’ theorem, the joint posterior distribution is proportional to the likelihood of the observed data multiplied by the prior distributions:

$$p(\mathbf{\Pi}, \mathbf{\Sigma} \mid \mathbf{Y}, \mathbf{X}) \propto p(\mathbf{Y} \mid \mathbf{X}, \mathbf{\Pi}, \mathbf{\Sigma})p(\mathbf{\Pi} \mid \mathbf{\Sigma})p(\mathbf{\Sigma}). \quad (4.16)$$

Following a non-informative Bayesian approach, a flat prior is assigned to the coefficient matrix,

$$p(\mathbf{\Pi} \mid \mathbf{\Sigma}) \propto 1, \quad (4.17)$$

while the covariance matrix is assigned the Jeffreys prior,

$$p(\mathbf{\Sigma}) \propto |\mathbf{\Sigma}|^{-(n+1)/2}. \quad (4.18)$$

Posterior simulation is carried out by drawing from the corresponding conditional posterior distributions of $\mathbf{\Pi}$ and $\mathbf{\Sigma}$. By repeating this procedure thousands of times, we

obtain posterior draws from the joint posterior distribution. A detailed derivation of the posterior distributions and a description of the Bayesian estimation procedure can be found in Maffei-Faccioli and Vella (2021).

4.1.2 Structural Identification

The reduced-form disturbances $\mathbf{u}_t$ are generally contemporaneously correlated. Therefore, a structural identification scheme is required in order to interpret the innovations as economically meaningful shocks. The relationship between the reduced-form disturbances and the structural disturbances is given by

$$\mathbf{u}_t = \mathbf{S}\boldsymbol{\varepsilon}_t, \quad (4.19)$$

where $\boldsymbol{\varepsilon}_t$ is an $n \times 1$ vector of structural disturbances satisfying

$$\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n). \quad (4.20)$$

Consequently, the covariance matrix of the reduced-form disturbances can be written as

$$\boldsymbol{\Sigma} = \mathbf{S}\mathbf{S}'. \quad (4.21)$$

Structural shocks are identified using a recursive Cholesky decomposition of $\boldsymbol{\Sigma}$, where $\mathbf{S}$ is the lower triangular matrix satisfying

$$\mathbf{S}\mathbf{S}' = \boldsymbol{\Sigma}. \quad (4.22)$$

The Cholesky decomposition is applied to the endogenous variables only. Accordingly, the exogenous COVID-19 dummy included in the extended 2007–2025 specification does not constitute an additional structural shock and does not affect the dimensionality or ordering of the structural innovation vector.

4.2 Data

The baseline specification has five endogenous variables: net immigration rate $NetImmgr_t$ ³; log Business Expectations Index⁴ $\log(BExp_t)$; log Consumer Confidence Index⁵ $\log(CConf_t)$; log Industrial Production Index⁶ $\log(IndPr_t)$; and registered unemployment rate⁷ $UnemplR_t$:

$$\mathbf{y}_t = \begin{bmatrix} NetImmgr_t \\ \log(BExp_t) \\ \log(CConf_t) \\ \log(IndPr_t) \\ UnemplR_t \end{bmatrix}. \quad (4.23)$$

So the baseline VAR has $n = 5$ endogenous variables. In the alternative specifications, we add one immigration sentiment indicator at a time, ordered last, giving six endogenous variables:

$$\mathbf{y}_t^{(s)} = \begin{bmatrix} NetImmgr_t \\ \log(BExp_t) \\ \log(CConf_t) \\ \log(IndPr_t) \\ UnemplR_t \\ S_{it} \end{bmatrix}, \quad i \in \{Government, Opposition\}, \quad (4.24)$$

where S_{it} is immigration sentiment for speaker group i .

We follow Maffei-Faccioli and Vella (2021) in choosing the macroeconomic variables. Consumer confidence and business expectations are forward-looking indicators, included because migration decisions may depend on expected, not just current, em-

³The net migration rate (expressed per 1,000 inhabitants) is constructed using data from the German Federal Statistical Office (Statistisches Bundesamt (Destatis) 2026b; Statistisches Bundesamt (Destatis) 2026a). Because official net migration balance data from Destatis are consistently available from 2008 onward (Table 12711-0011), the pre-2008 historical observations (covering 2006 and 2007) are extended using the underlying monthly administrative migration series from Maffei-Faccioli and Vella (2021).

⁴Data regarding the business climate index are sourced from the Ifo Institute (Ifo Institute 2026).

⁵Data on consumer confidence are obtained from the Organisation for Economic Co-operation and Development (OECD 2026).

⁶Data on industrial production are obtained from the Deutsche Bundesbank statistical time-series database (Deutsche Bundesbank 2026).

⁷The unemployment rate is the share of registered unemployed in the labor force, where the labor force is employed German residents (Destatis) plus registered unemployed (Bundesagentur für Arbeit).

ployment and income conditions; adding forward-looking variables also helps limit the risk of non-fundamentalness (Forni and Gambetti 2014).

The Consumer Confidence Index comes from household assessments of their financial situation, the general economy, employment prospects, and savings, sourced from the Organisation for Economic Co-operation and Development (OECD). The Business Expectations Index comes from monthly ifo Institute surveys of firms in manufacturing, services, trade, and construction, on their expectations for the next six months. Industrial production captures overall economic activity, and registered unemployment lets us directly examine the link between labour-market conditions and immigration.

All log series are multiplied by 100, and the data are monthly. Our analysis reconstructs and extends the dataset from Maffei-Faccioli and Vella (2021).

We estimate three VAR models over three sample periods. First, 2006–2019, matching the original study, so our reconstructed data and estimates can be compared directly against Maffei-Faccioli and Vella (2021) as a validation check. Second, 2007–2019 on our reconstructed data, motivated by a large discrepancy we found between the original study’s consumer confidence series and the series from its stated source (Figure 10); this lets us check whether results are sensitive to our reconstructed series while staying close to the original sample. Third, the extended 2007–2025 period, adding the COVID-19 pandemic and its aftermath, with a binary COVID-19 dummy as an exogenous control (Eq. 4.5) to keep pandemic-driven disruption from being wrongly attributed to the endogenous relationships between immigration, the economy, and political sentiment. This dummy appears only in the 2007–2025 specification and is excluded from both the endogenous vector and the structural shocks.

4.3 Identification

We identify immigration and sentiment shocks with a recursive (Cholesky) ordering, following (Maffei-Faccioli and Vella 2021) for the migration block and extending it by placing our new Bundestag immigration sentiment measure last, as in Eq. 4.24. We report three specifications, one per sentiment measure ($i \in \{\textit{Government}, \textit{Opposition}\}$); only the sentiment series changes across them, so the migration and macro block ordering stays fixed, and any difference in results can be attributed to the sentiment measure itself, not the identification scheme.

Data Comparison: Extended Dataset vs. Base Study Data

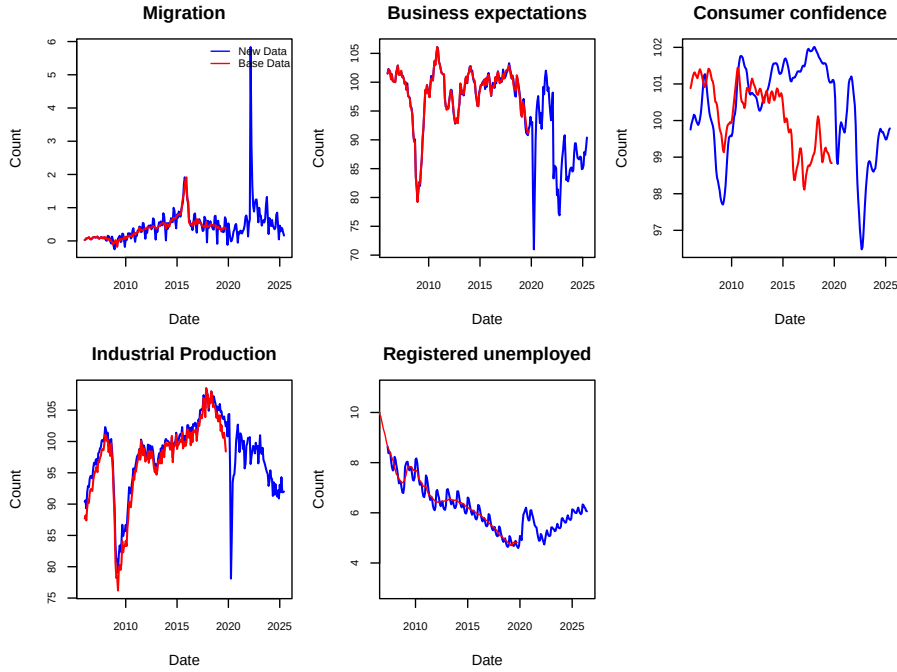


Figure 10: Data Comparison: Extended Dataset vs. Base Study Data. Five time-series charts (2010–2025) comparing New Data (blue) and Base Data (red) for Migration, Business expectations, Consumer confidence, Industrial Production, and Registered unemployed.

Ordering the migration variable. Arrivals are ordered first, assuming German economic conditions do not affect migration within the same month: as (Maffei-Faccioli and Vella 2021) argue, deciding to migrate, getting authorization, and arriving takes longer than a month, so a same-month economic shock is unlikely to drive contemporaneous migration.

Ordering the macroeconomic block. We order $BExp_t$, $CConf_t$, $IndPr_t$, $UnemplR_t$, following (Maffei-Faccioli and Vella 2021): the forward-looking survey measures (expectations, confidence) come before the realized ("harder") indicators (production, unemployment). This internal ordering is not critical to our results, since both survey and realized variables sit between arrivals and sentiment in every specification; re-ordering within this block does not affect the sentiment ordering or the identification of the sentiment shock, our main object of interest. We test sensitivity to this choice in Section 4.5.

Ordering the sentiment variable. Sentiment is ordered last because parliamentary immigration rhetoric is likely to respond contemporaneously to prevailing economic and social conditions, while its effects on migration and macroeconomic outcomes are expected to operate with a delay through public discourse and social amplification (Abdelatif and Andriantomanga 2026; Rakers 2023). The recursive identification therefore assumes that macroeconomic and migration variables may affect sentiment within the month, but sentiment does not affect them contemporaneously. This is an identifying assumption, which we assess through alternative recursive orderings in Section 4.5.

Predicted signs. This ordering implies directional predictions we can check against our results. If sentiment genuinely carries information about future immigration and economic conditions, a one-standard-deviation rise in immigration sentiment should, over the following one to two years: (i) raise arrivals, business expectations, and industrial production; (ii) lower consumer confidence in the short run, consistent with natives reading more open rhetoric as a signal of stronger labour-market competition; and (iii) lower unemployment over the medium term. Table 6 summarizes these predictions.

Variable	Sentiment shock
Arrivals	+
Business expectations	+
Consumer confidence	– (short run)
Industrial production	+
Unemployment	– (medium run)

Table 6: Predicted sign of the response to a one-standard-deviation sentiment shock, under the identifying assumptions in this section.

4.4 Results

This section presents the estimation results for several Bayesian VAR specifications, using different datasets and sample periods, and interprets the findings in light of the existing literature. We proceed in three steps. First, we replicate the baseline results of (Maffei-Faccioli and Vella 2021) using their original dataset and extend their model with three measures of immigration-related political sentiment (government, opposition, Bundestag). Second, we re-estimate the same specifications on a reconstructed

dataset covering an extended sample period through June 2025. Third, we summarize what is robust across specifications and datasets and what is not.

4.4.1 Original Dataset from (Maffei-Faccioli and Vella 2021)

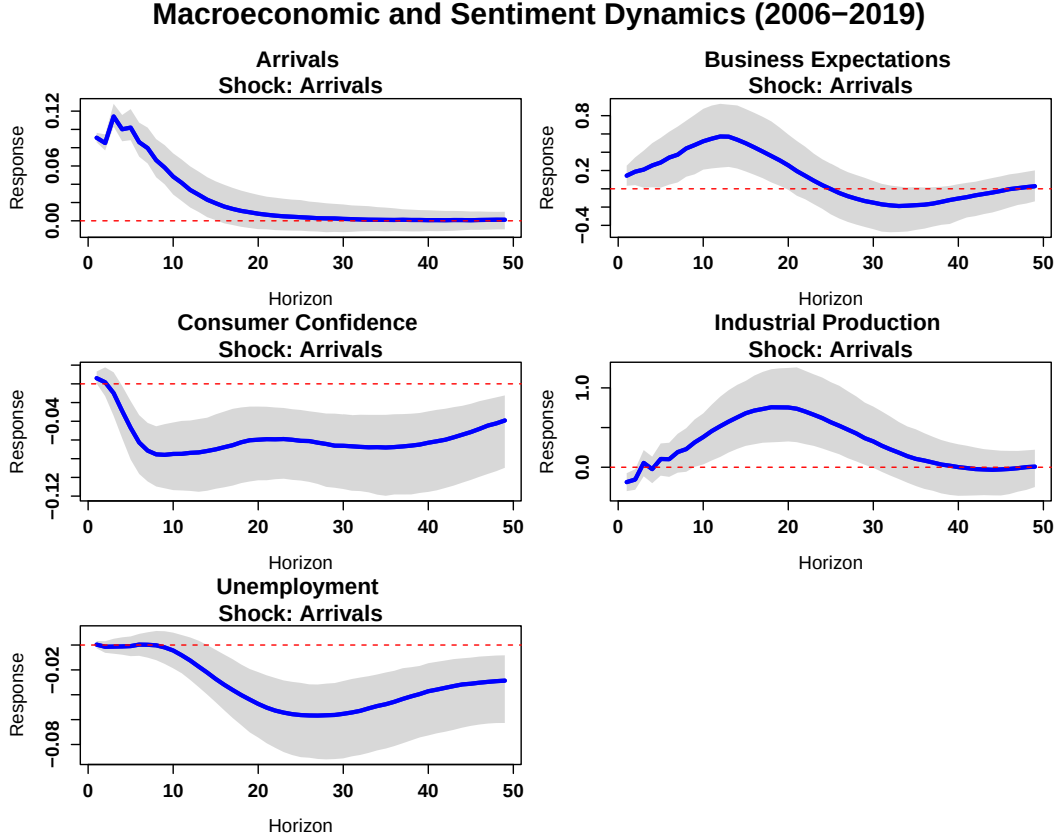


Figure 11: Impulse response functions to a one-standard-deviation net migration shock. The plots present the first main result of (Maffei-Faccioli and Vella 2021), showing the responses of the endogenous variables to a positive shock to net migration. The impulse responses are obtained from a recursively identified Bayesian VAR model with three lags and Normal–Wishart priors. The shaded areas represent the 68% credible intervals based on posterior draws.

We first reproduce the main results of (Maffei-Faccioli and Vella 2021) using their original dataset. Our results are broadly consistent with theirs (Figure 11). A positive shock to immigration leads to a short-term increase in business expectations and industrial production, while consumer confidence declines significantly. The original authors interpret this decline as reflecting concerns among the German population about increased labour-market competition; we return to this interpretation, and its limits, below. Unemployment also decreases following the immigration shock, with the response becoming significant approximately nine months after impact.

Extending the baseline model with sentiment measures. Figures 12, 13, and 14 show the responses of the endogenous variables to a positive shock in immigration sentiment, as reflected in speeches by government members, opposition members, and the Bundestag as a whole, respectively.

Across all three shocks, business expectations and industrial production decline significantly over comparable horizons. Unemployment rises significantly following opposition and Bundestag sentiment shocks (see 13, 14), while the response to a government sentiment shock is comparatively weaker (see 12). Consumer confidence declines significantly only after Bundestag and government sentiment shocks (see 12, 14); under an opposition sentiment shock, its response remains close to zero. Arrivals decline slightly following all three shocks, but the response is statistically negligible throughout.

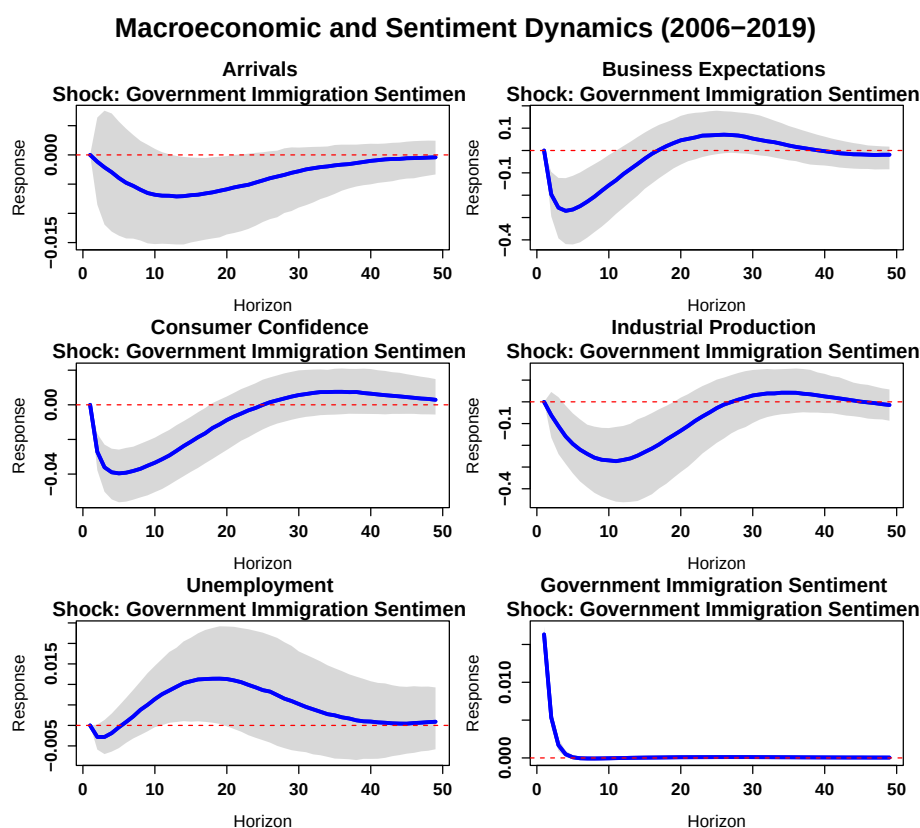


Figure 12: Macroeconomic and Sentiment Dynamics (2006–2019). The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches by members of the governing parties (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with one lag and Normal–Wishart priors.

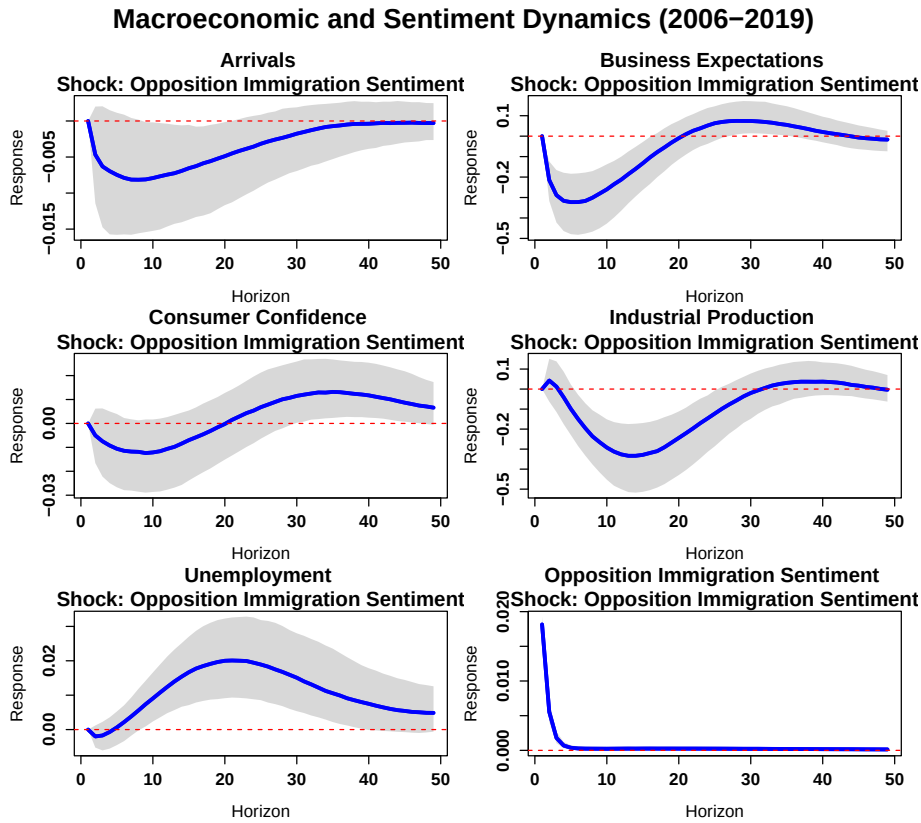


Figure 13: Macroeconomic and Sentiment Dynamics (2006–2019). The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches by members of the opposition parties (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with one lag and Normal–Wishart priors.

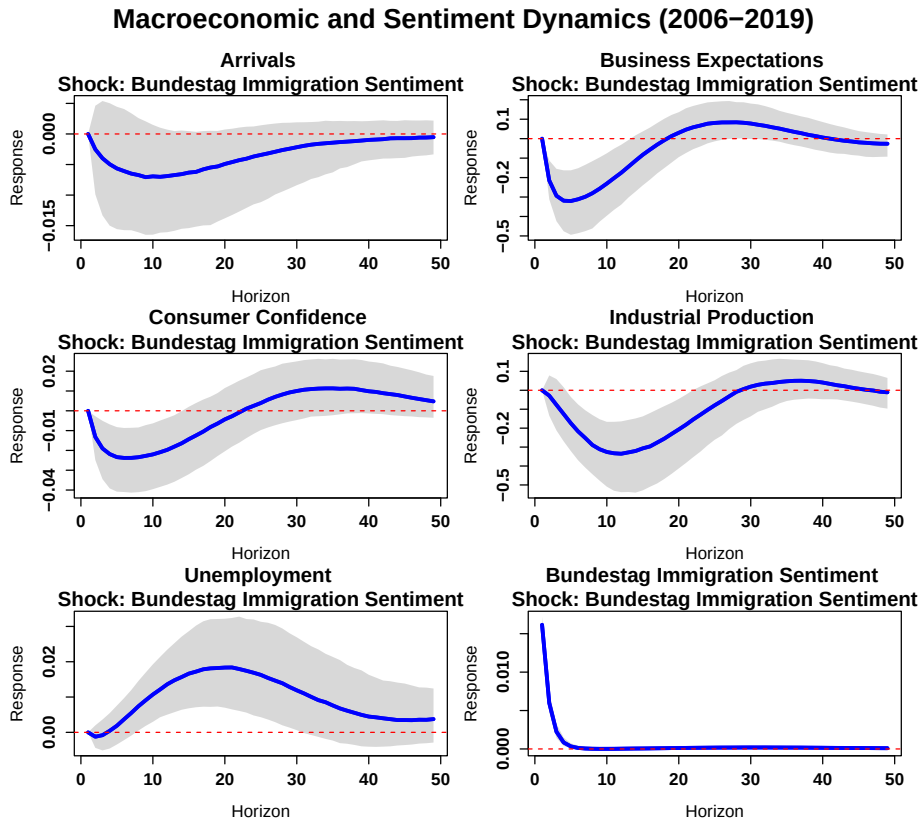


Figure 14: Macroeconomic and Sentiment Dynamics (2006–2019). The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches of all members of the Bundestag (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with one lag and Normal–Wishart priors.

4.4.2 Extension of the Maffei et al. (2021) Dataset to June 2025

We now turn to the reconstructed dataset, which extends the sample period to June 2025. As discussed above, the reconstructed consumer confidence series differs from the one used in the original study (see Figure 10), motivating separate estimation over the pre-pandemic window 2007–2019 and the full extended window 2007–2025, with a COVID-19 exogenous dummy included in the latter.

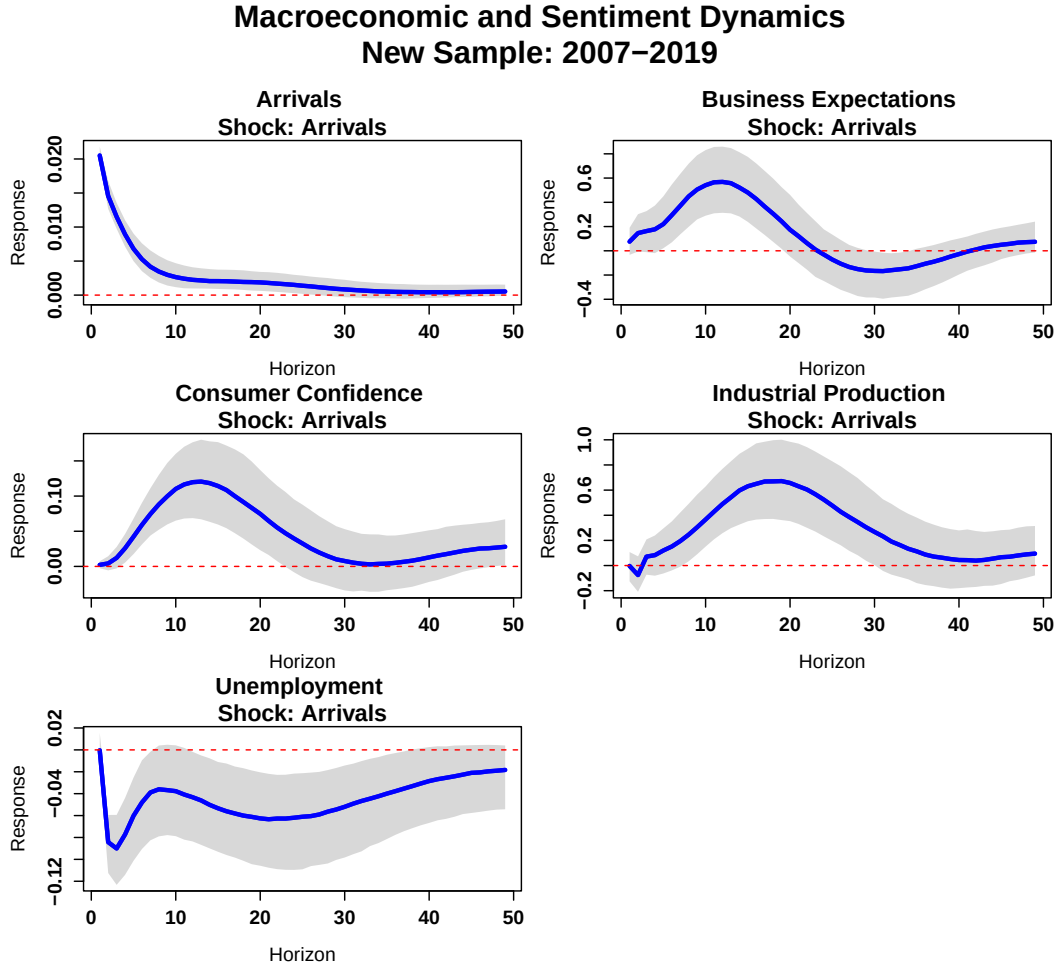


Figure 15: Macroeconomic and Sentiment Dynamics, New Sample: 2007–2019. Impulse response functions to a one-standard-deviation arrival shock. Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with two lags and a diffuse (flat-Jeffreys) prior.

Figures 15 and 16 report the responses to an arrival shock, partially replicating the findings of the original study (Figure 11). The decline in unemployment and the rise in industrial production appear across all specifications, though with varying magnitudes. Business expectations and consumer confidence are significant in the original study and the 2007–2019 sample, but lose significance in the extended 2007–2025

sample. Additionally, the reconstructed sample yields a positive consumer confidence response, unlike the negative response in the original study. This difference might stem from two factors: first, discrepancies in the underlying OECD consumer confidence data series for the overlapping period; second, the structural multi-crisis environment of 2020–2025 (comprising the pandemic, energy crisis, inflation, and economic stagnation) which alters macroeconomic dynamics.

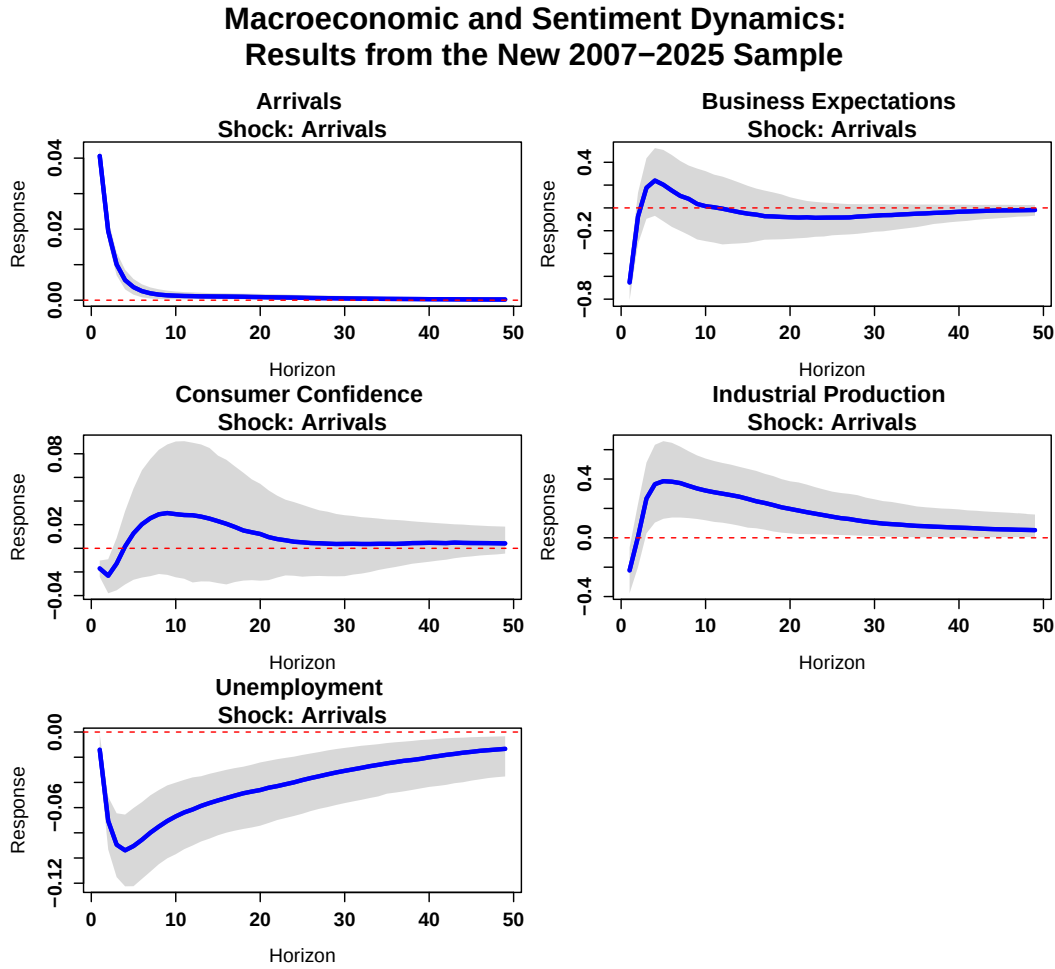


Figure 16: Macroeconomic and Sentiment Dynamics, New Sample: 2007–2025. Impulse response functions to a one-standard-deviation arrival shock. Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with two lags and a diffuse (flat-Jeffreys) prior, including a COVID-19 exogenous control dummy for observations between March 2020 and October 2021.

In short, the reconstructed dataset yields a significant relationship between immigration shocks and macroeconomic variables, although the strength and significance of this relationship depend on the sample period considered. This sensitivity is likely attributable, at least in part, to the dataset reconstruction differences discussed in Fig-

ure 10.

Reconstructed dataset, 2007–2019 window. Figures 17, 18, and 19 show the responses of the endogenous variables to sentiment shocks computed from speeches by government members, opposition members, and the Bundestag as a whole, respectively. The overall pattern is broadly consistent with the original sample (Figures 12–14): business expectations and consumer confidence decline sharply following each shock. The rise in unemployment, however, is markedly stronger and more persistent here than in the original sample (see figures 12, 13, 14), and this is especially pronounced under the opposition and Bundestag sentiment shocks (see figure 18), where the response is both larger in magnitude and slower to revert than under the government shock.

Arrivals respond negatively and significantly to an opposition sentiment shock, while the response is comparatively small and less clearly significant for the government and Bundestag shocks (see figures 17, 19).

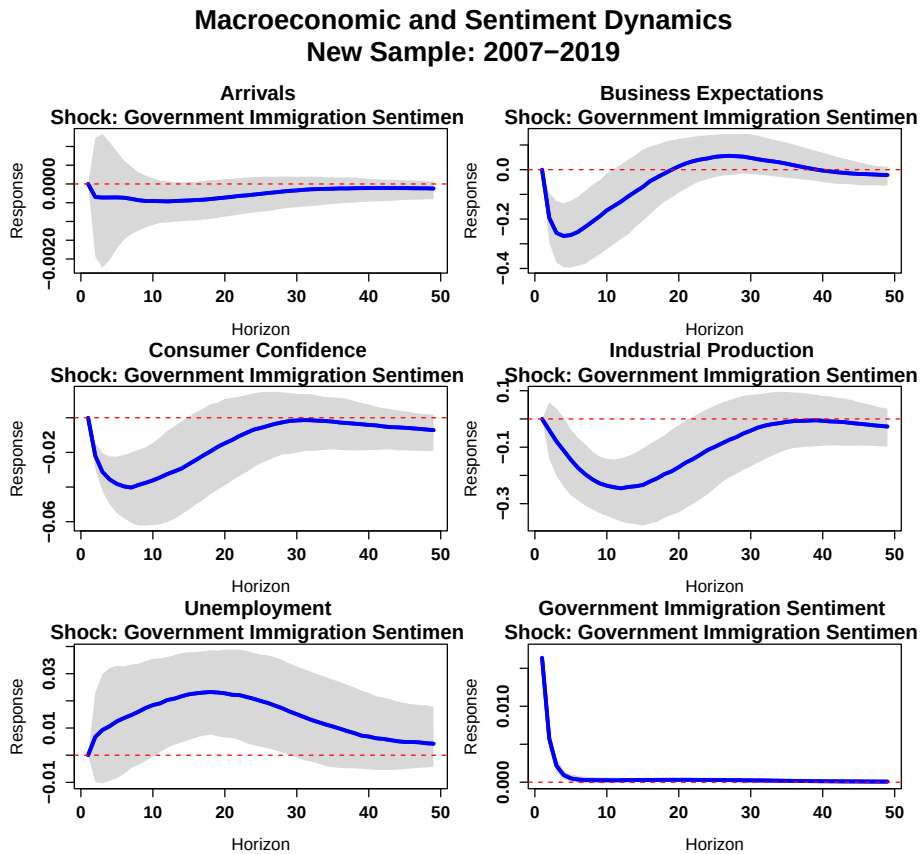


Figure 17: Macroeconomic and Sentiment Dynamics, New Sample: 2007–2019. The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches by members of the governing parties (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with one lag and Normal–Wishart priors.

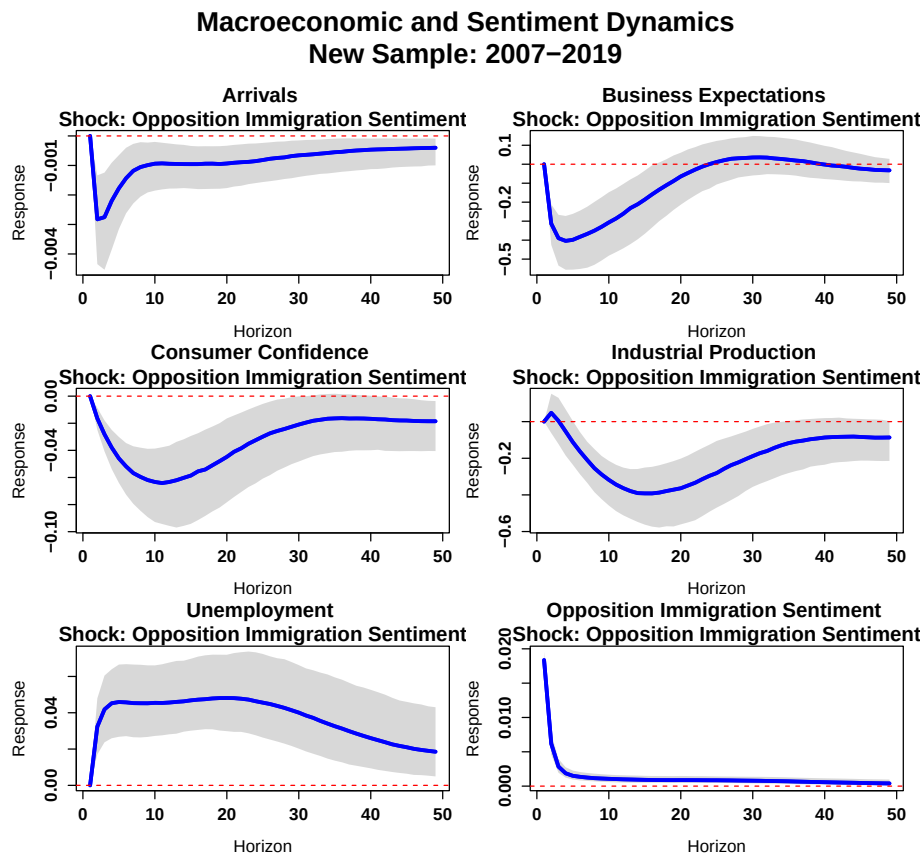


Figure 18: Macroeconomic and Sentiment Dynamics, New Sample: 2007–2019. The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches by members of the opposition parties (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with one lag and Normal–Wishart priors.

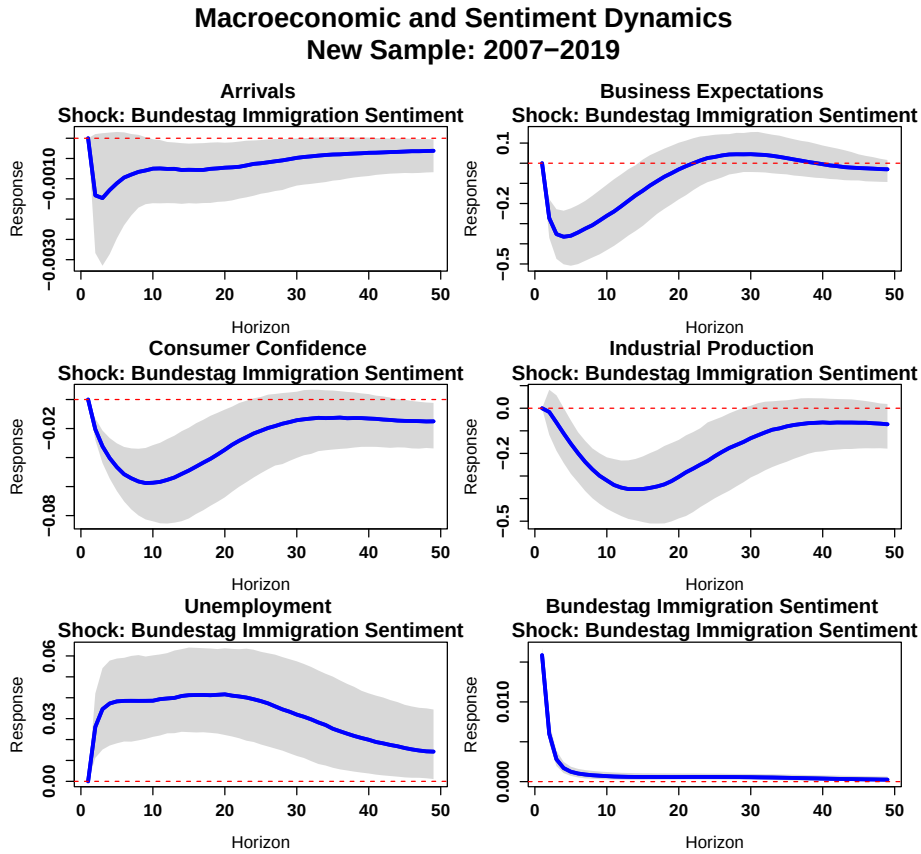


Figure 19: Macroeconomic and Sentiment Dynamics, New Sample: 2007–2019. The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches of all members of the Bundestag (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with one lag and Normal–Wishart priors.

Extended sample: 2007–2025. Figures 20, 21, and 22 report the corresponding impulse responses. The pattern differs markedly from the government shock, on one hand, and the opposition and Bundestag shocks, on the other. Following a government sentiment shock, most endogenous variables respond weakly and without clear significance (see figure 20), with the exception of unemployment, which remains positive throughout the horizon. Following opposition and Bundestag sentiment shocks, unemployment, business expectations, and consumer confidence all rise significantly, while arrivals decline significantly; industrial production, by contrast, shows no statistically discernible response (see figures 21, 22).

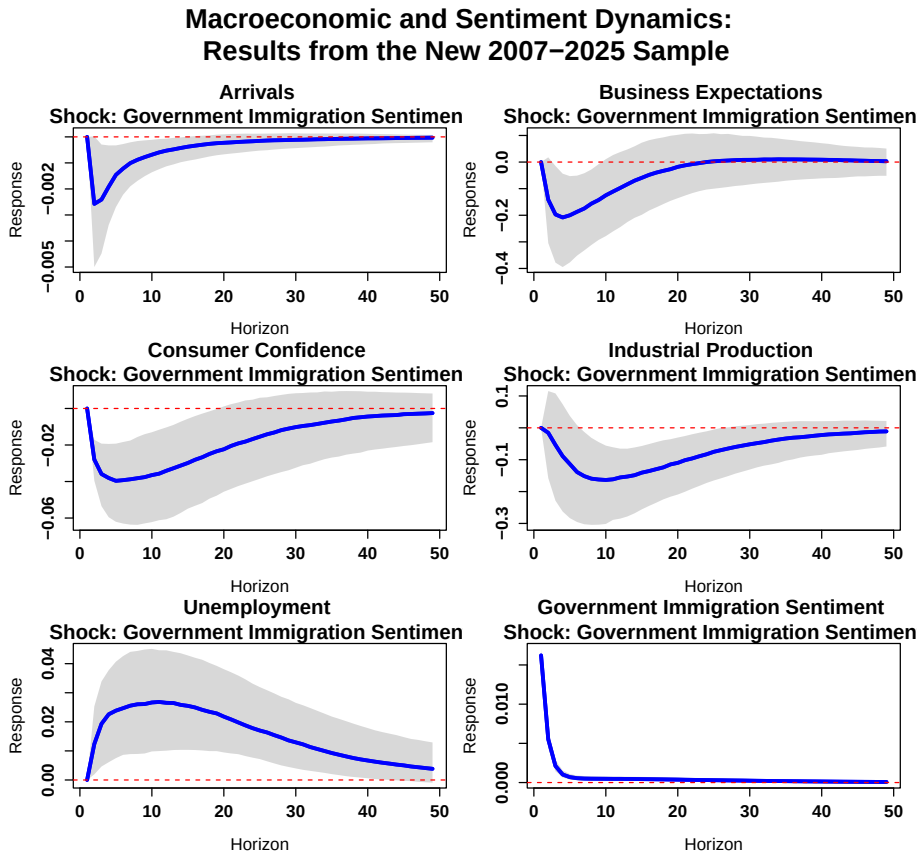


Figure 20: Macroeconomic and Sentiment Dynamics, New Sample: 2007–2025. The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches by members of the governing parties (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with two lags and Normal–Wishart priors.

Taken together, sentiments are associated with subsequent macroeconomic dynamics. However, we defer a detailed economic interpretation of these findings to the robustness section (4.5). This postponement is necessary due to limitations identified

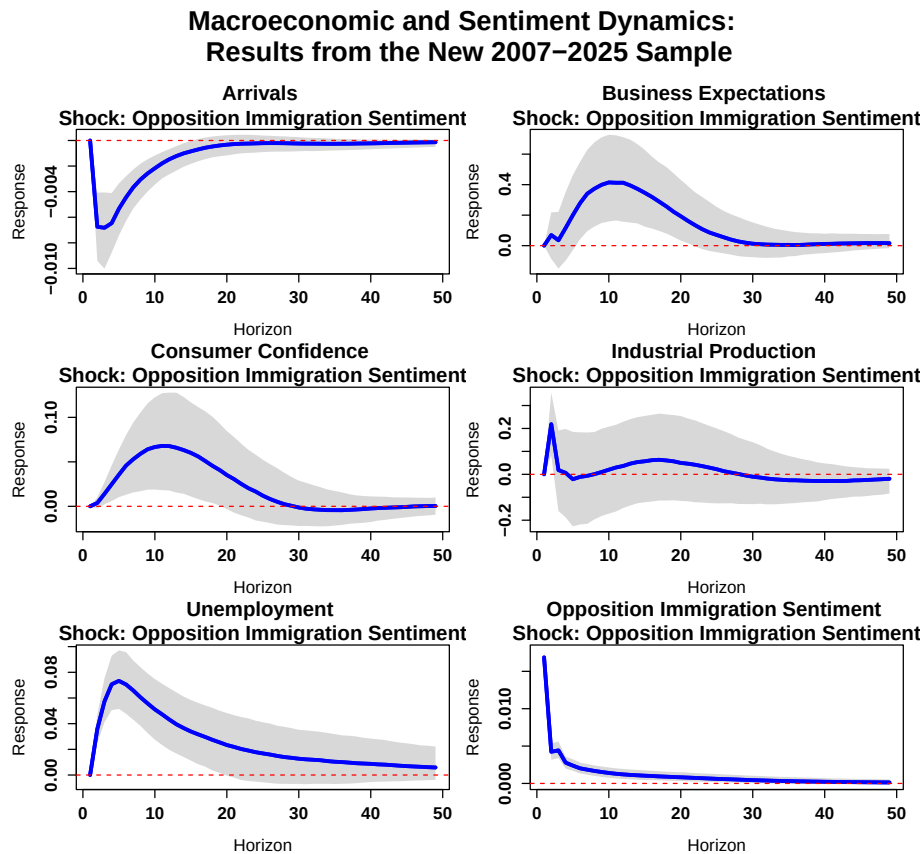


Figure 21: Macroeconomic and Sentiment Dynamics, New Sample: 2007–2025. The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches by members of the opposition parties (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with two lags and Normal–Wishart priors.

in our Doc2Vec model—specifically, its sensitivity to negative rhetoric, which can bias the computation of immigration sentiment within Bundestag speeches.

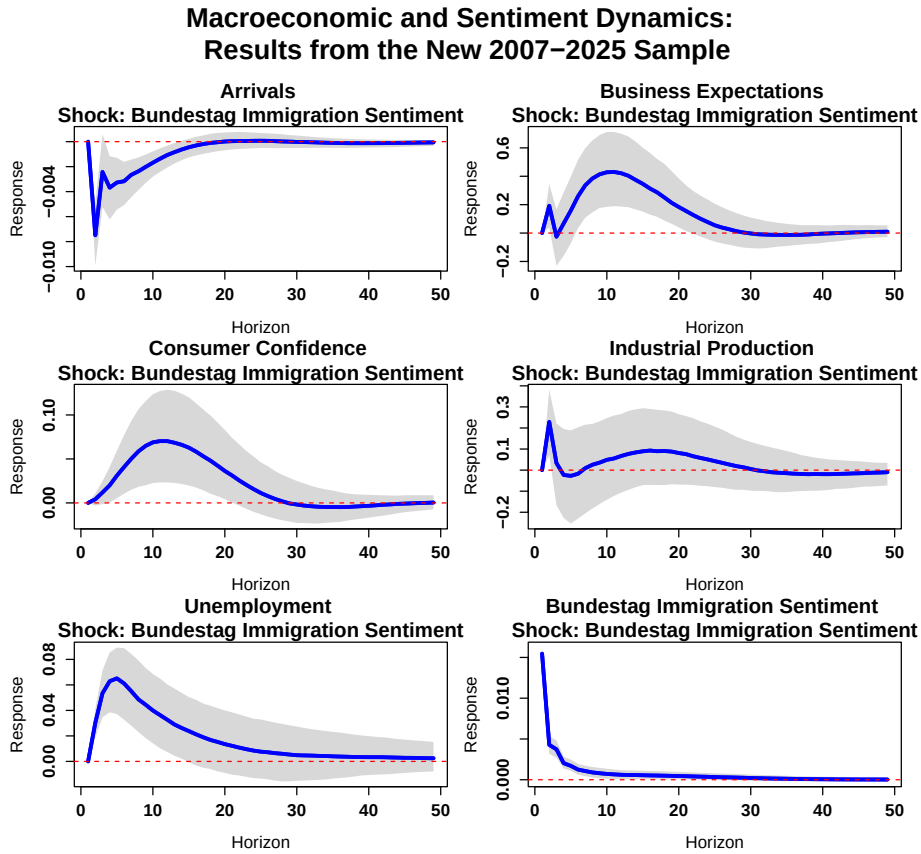


Figure 22: Macroeconomic and Sentiment Dynamics, New Sample: 2007–2025. The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches of all members of the Bundestag (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with two lags and Normal–Wishart priors.

4.5 Robustness

As a further robustness check on the identification scheme, we re-estimate the opposition-sentiment specification on the pre-crisis subsample 2007–2014 under three alternative recursive orderings. Restricting the sample to this window additionally allows us to assess whether the sentiment–unemployment relationship documented in Section 4.4.2 is already present before the 2015–2016 refugee episode, rather than being generated primarily by it.

Specification 1: baseline ordering. The first ordering follows the baseline scheme of Equation 4.25, with arrivals ordered first, the macroeconomic block ordered as business expectations, consumer confidence, and industrial production, and opposition sentiment ordered last:

$$\mathbf{y}_t = [\text{Arrivals}_t, \text{BusExp}_t, \text{ConsConf}_t, \text{IndProd}_t, \text{Unemp}_t, \text{Sent}_t^{\text{Opp}}]' . \quad (4.25)$$

Figure 23 reports the resulting impulse responses. Arrivals decline on impact and remain significant for approximately 7 months following the shock, while unemployment rises on impact and remains significant for approximately 10–15 months. This indicates that the effect of opposition immigration sentiment on arrivals and unemployment is already present in the pre-crisis window, although it is comparatively short-lived relative to the full 2007–2025 sample.

Specification 2: reordered macroeconomic block. The second ordering places the two realized indicators, industrial production and unemployment, ahead of the two survey-based expectations variables, while keeping arrivals first and sentiment last:

$$\mathbf{y}_t = [\text{Arrivals}_t, \text{IndProd}_t, \text{Unemp}_t, \text{BusExp}_t, \text{ConsConf}_t, \text{Sent}_t^{\text{Opp}}]' . \quad (4.26)$$

Figure 24 shows results essentially unchanged from Specification 1: the responses of arrivals and unemployment remain of similar magnitude and significance, indicating that the results are robust to this reordering.

Specification 3: sentiment ordered second. The third ordering moves opposition sentiment to the second position, immediately after arrivals and ahead of the entire

macroeconomic block, thereby allowing sentiment to affect all four macroeconomic variables contemporaneously:

$$\mathbf{y}_t = [\text{Arrivals}_t, \text{Sent}_t^{\text{Opp}}, \text{IndProd}_t, \text{Unemp}_t, \text{BusExp}_t, \text{ConsConf}_t]'. \quad (4.27)$$

Figure 25 shows a similar pattern to Specifications 1 and 2, with arrivals and unemployment again responding significantly and over comparable horizons. Overall, the three orderings deliver consistent conclusions, suggesting that the documented responses are not highly sensitive to the alternative recursive orderings considered.

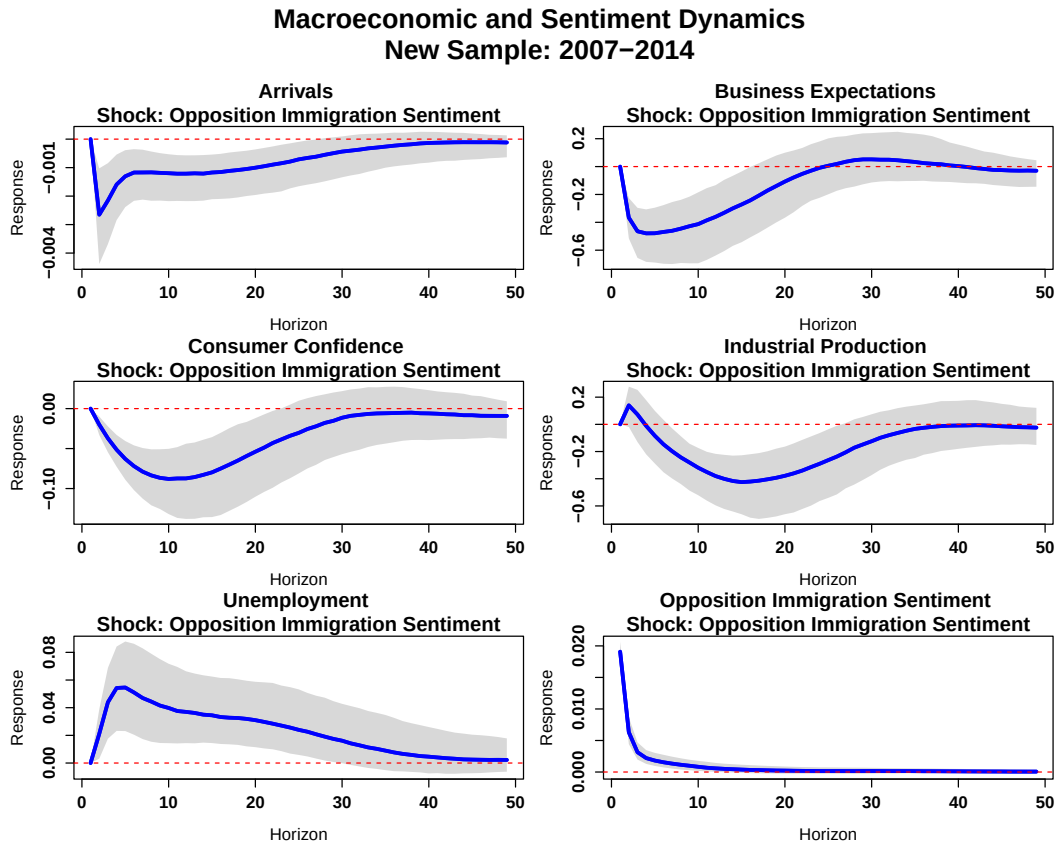


Figure 23: **Robustness check, Specification 1 (baseline ordering), sample 2007–2014.** The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches by members of the opposition parties (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with one lag and Normal–Wishart priors.

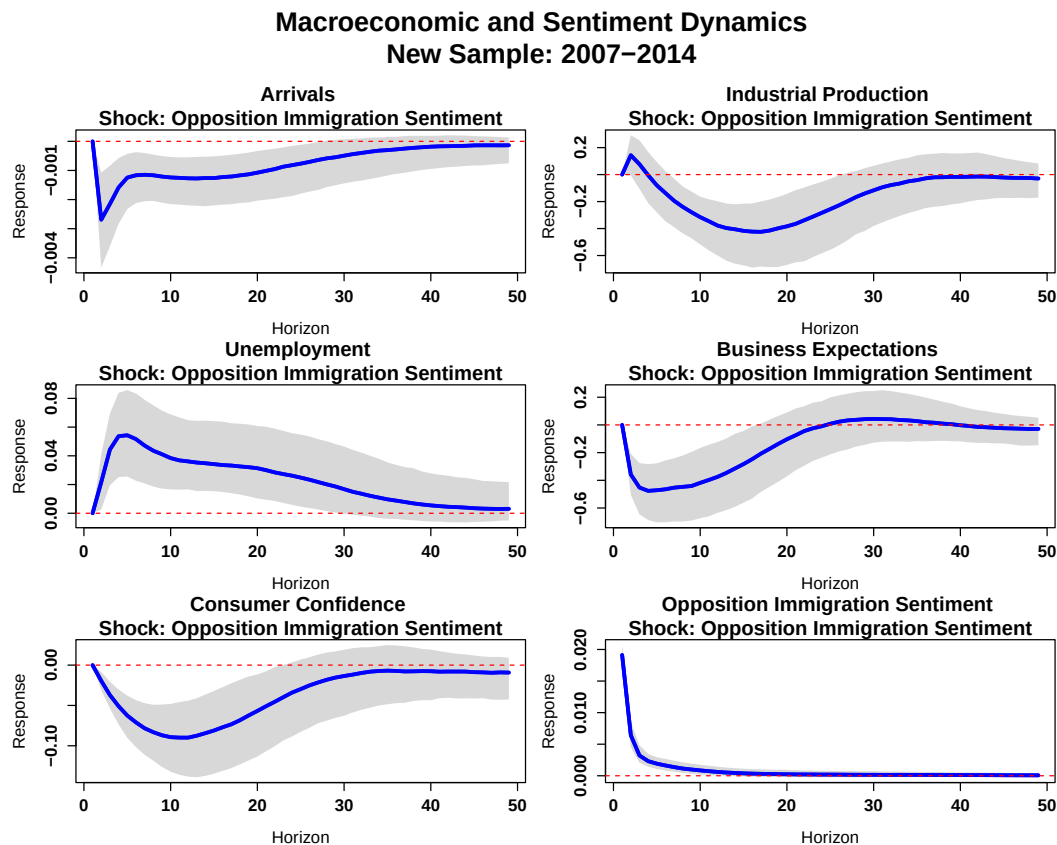


Figure 24: **Robustness check, Specification 2 (reordered macroeconomic block), sample 2007–2014.** The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches by members of the opposition parties (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with one lag and Normal–Wishart priors.

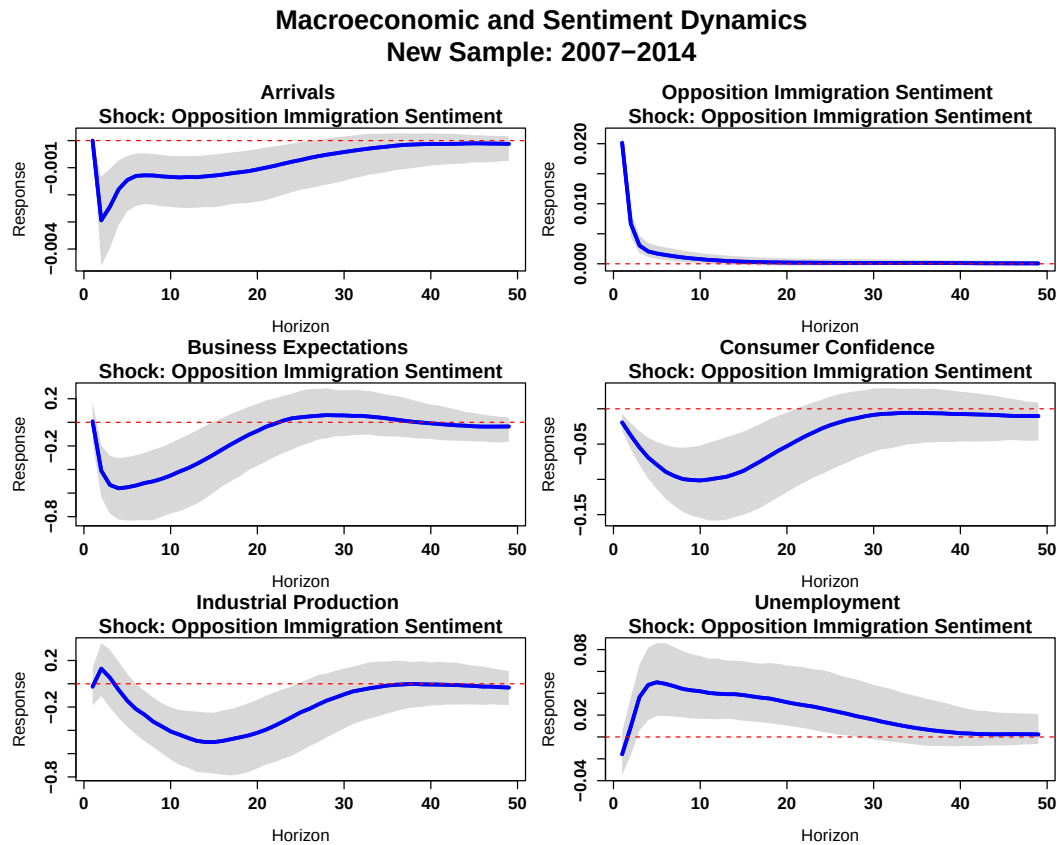


Figure 25: **Robustness check, Specification 3 (sentiment ordered second), sample 2007–2014.** The figure shows the responses of the endogenous variables to an increase in immigration sentiment, as reflected in speeches by members of the opposition parties (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with one lag and Normal–Wishart priors.

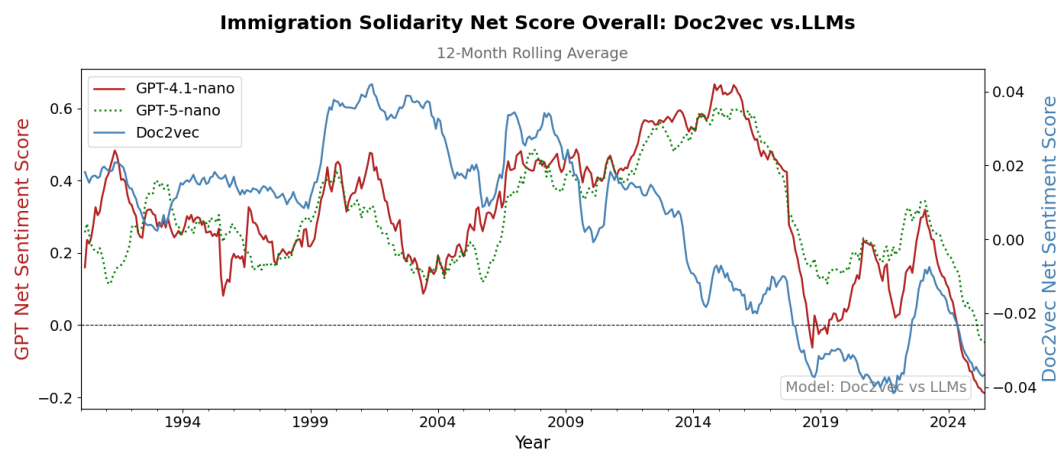


Figure 26: 12-month rolling average of the immigration solidarity net score comparing Doc2vec and LLM.

Specification 4: alternative sentiment measure LLM. The three specifications above test robustness to the recursive ordering, keeping the sentiment measure fixed. A separate concern, raised in Section 3.2, is that Doc2Vec struggles with negation. Manual inspection found several misclassified speeches, and even after the correction in Section 3.2, average sentiment for some parties (CDU/CSU, Bündnis 90/Die Grünen, Die Linke) still looked implausible. As a check on the sentiment measure itself, we re-estimate immigration sentiment using large language models (GPT-4.1-nano and GPT-5-nano), following Kostikova et al. (2024). Each migration-related sentence is classified into solidarity or anti-solidarity subtypes, and we build a monthly net-solidarity score the same way as the dictionary-based measure, separately for government and opposition speakers. The full prompt is in Appendix 9.

Figures 27, 28, and 29 show that the LLM-based measure ranks parties in a way that better matches their known public rhetoric than the uncorrected dictionary measure. This supports our earlier suspicion that some of the discrepancies in Section 3.2 came from negation-related misclassification.

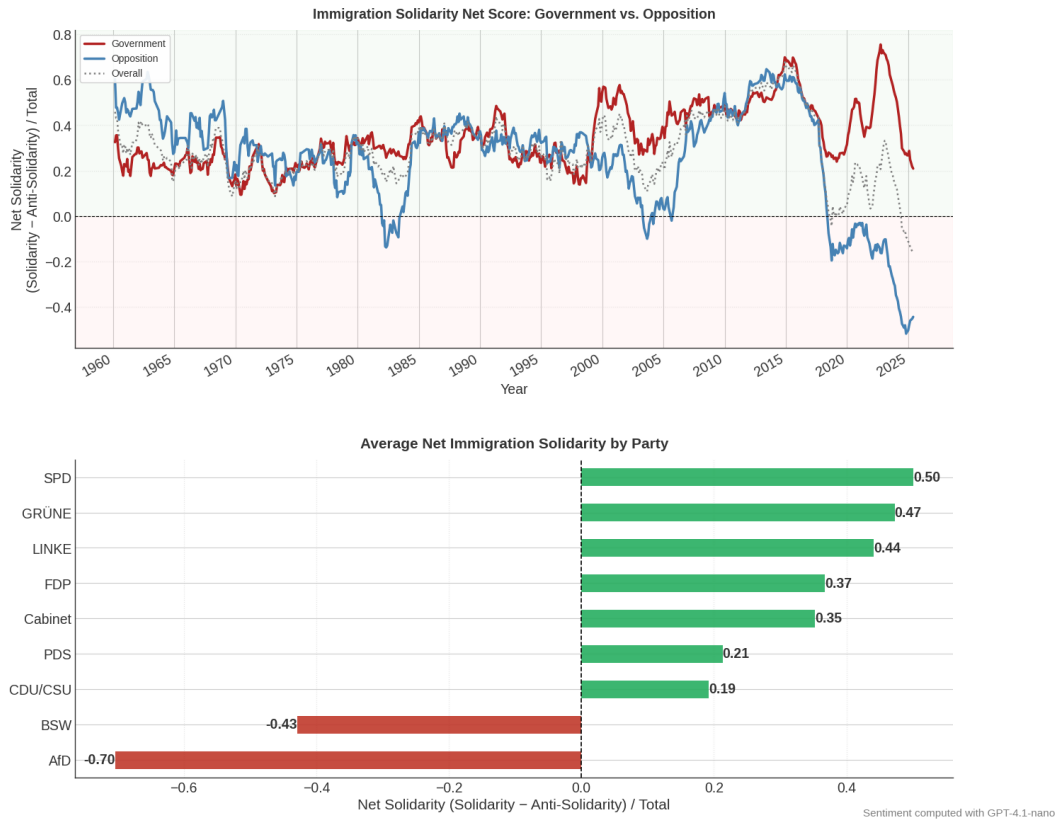


Figure 27: Average Net Immigration Solidarity by Party (GPT-4.1-nano).

We re-estimate the baseline VAR using the LLM-based opposition-sentiment series

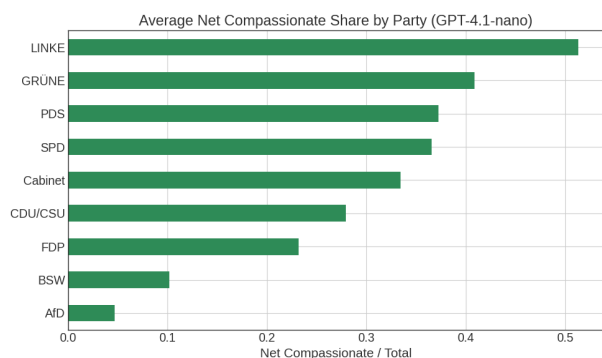


Figure 28: Average Net Compassionate Solidarity by Party (GPT-4.1-nano).

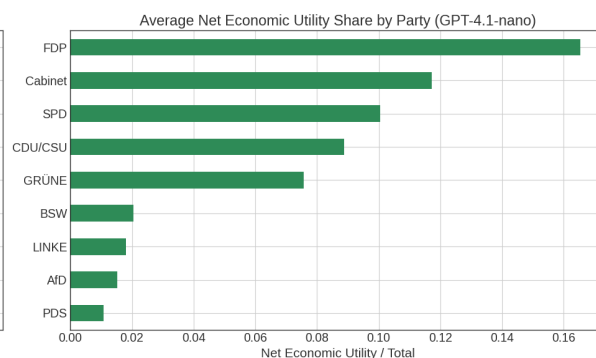


Figure 29: Average Net Economic-Utility Solidarity by Party (GPT-4.1-nano).

over three windows: 2007–2014, 2007–2019, and 2007–2025.

Figures 30, 31, and 32 show that opposition sentiment raises business expectations, consumer confidence, and industrial production. Industrial production does not react right away: it dips first, then turns positive. This dip lasts about two months in the 2007–2014 sample, six months in 2007–2019, and ten months in 2007–2025.

This pattern differs from the Doc2Vec results for 2007–2014 and 2007–2019, where business expectations, consumer confidence, and industrial production all **decline** after a sentiment shock. The likely reason is that the two sentiment series behave differently between 2011 and 2019: the Doc2Vec series drops sharply in this period, while the LLM series first rises and then falls sharply after 2014 (see Figure 26). In the 2007–2025 period, the two sentiment series follow a broadly similar overall trend, which is why their impulse responses end up looking similar in shape despite differing in level. Unemployment rises after the shock, most clearly in the 2007–2025 sample, where the response stays significant for about one and a half years.

Migration arrivals behave differently across the three windows: the response is negligible in 2007–2014, significantly positive for over ten months in 2007–2019, and negative before flattening out in 2007–2025. In other words, opposition pro-immigration rhetoric had no clear effect on arrivals before 2015, a positive effect afterward — plausibly linked to the 2015–2016 refugee crisis — and a negative effect once the sample is extended to 2025, which may reflect stricter migration policy and more restrictive public rhetoric in recent years.

Overall, the LLM-based sentiment measure (GPT-4.1-nano) offers a more context-sensitive way to capture the tone of the immigration debate, as shown by Figures 27,

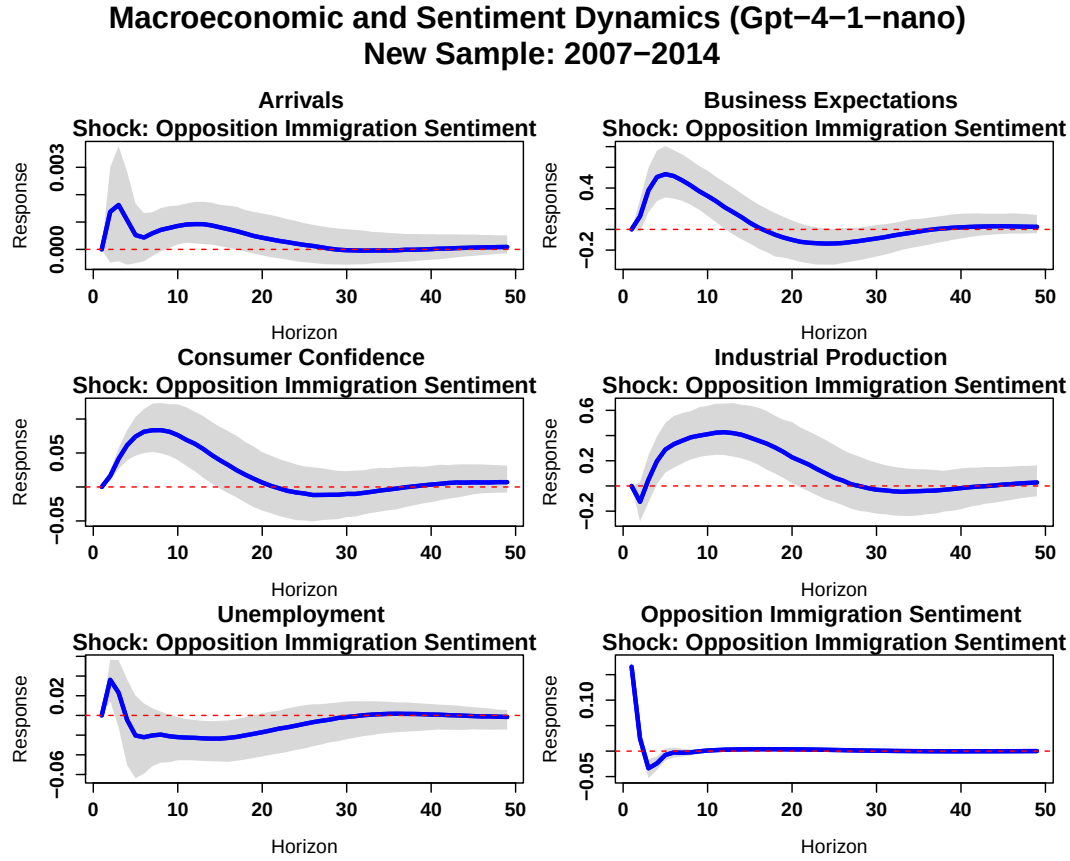


Figure 30: Robustness check, LLM-based sentiment measure, sample 2007–2014. The figure shows the responses of the endogenous variables to an increase in opposition immigration sentiment, constructed using GPT-4.1-nano following Kostikova et al. (2024) (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with two lags and Normal–Wishart priors.

28, and 29 and by the impulse responses above, consistent with Kostikova et al. 2024, who report a macro-F1 of 0.42 for GPT-4 on similar migration-related classification, against a human upper bound of 0.56. The differences from our dictionary-based measure are consistent with Doc2Vec’s known weakness in handling negation. Finally, the fact that the impulse-response pattern is stable across samples suggests that immigration sentiment mainly works through expectation channels — business expectations and consumer confidence — while unemployment and arrivals respond more to the broader macroeconomic conditions of each period.

Figure 33 shows the forecast error variance decomposition: the opposition immigration sentiment shock explains about 30% of monthly fluctuations in business expectations and consumer confidence after 10 to 15 months, but only about 5% of the variance in unemployment after around 5 months.

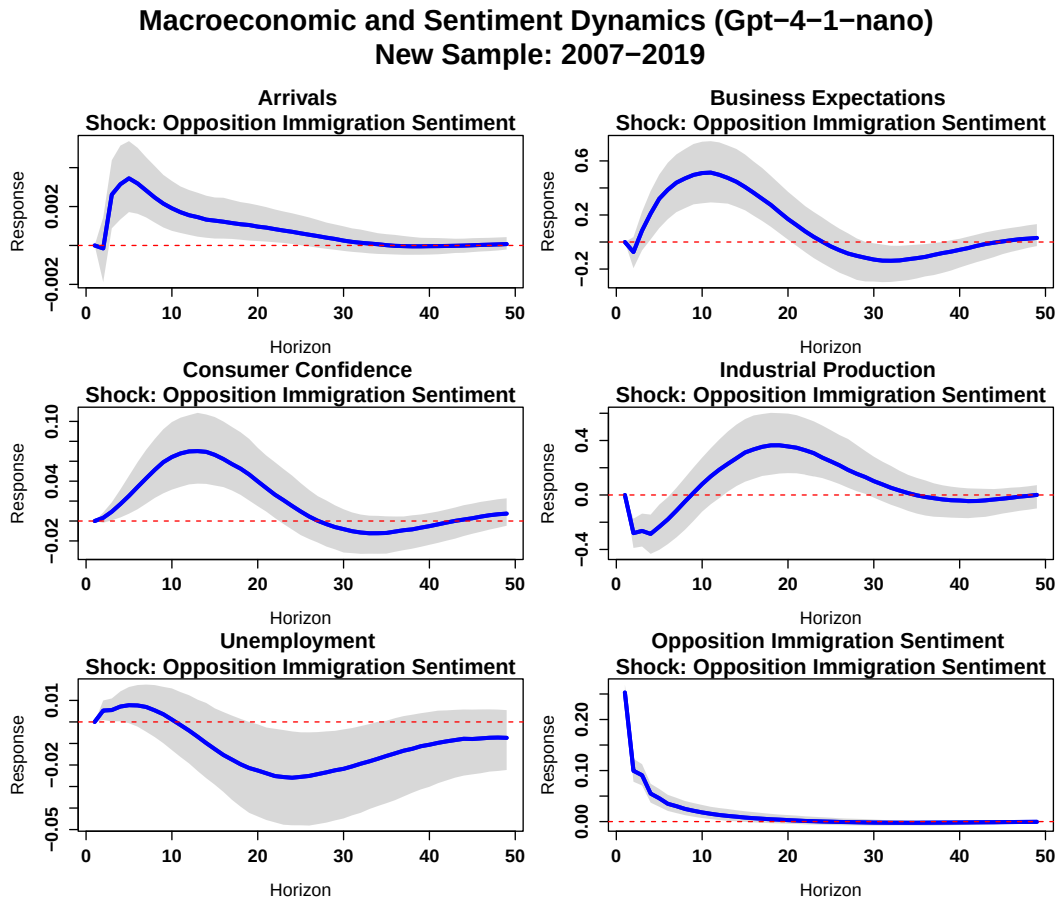


Figure 31: Robustness check, LLM-based sentiment measure, sample 2007–2019. The figure shows the responses of the endogenous variables to an increase in opposition immigration sentiment, constructed using GPT-4.1-nano following Kostikova et al. (2024) (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with two lags and Normal–Wishart priors.

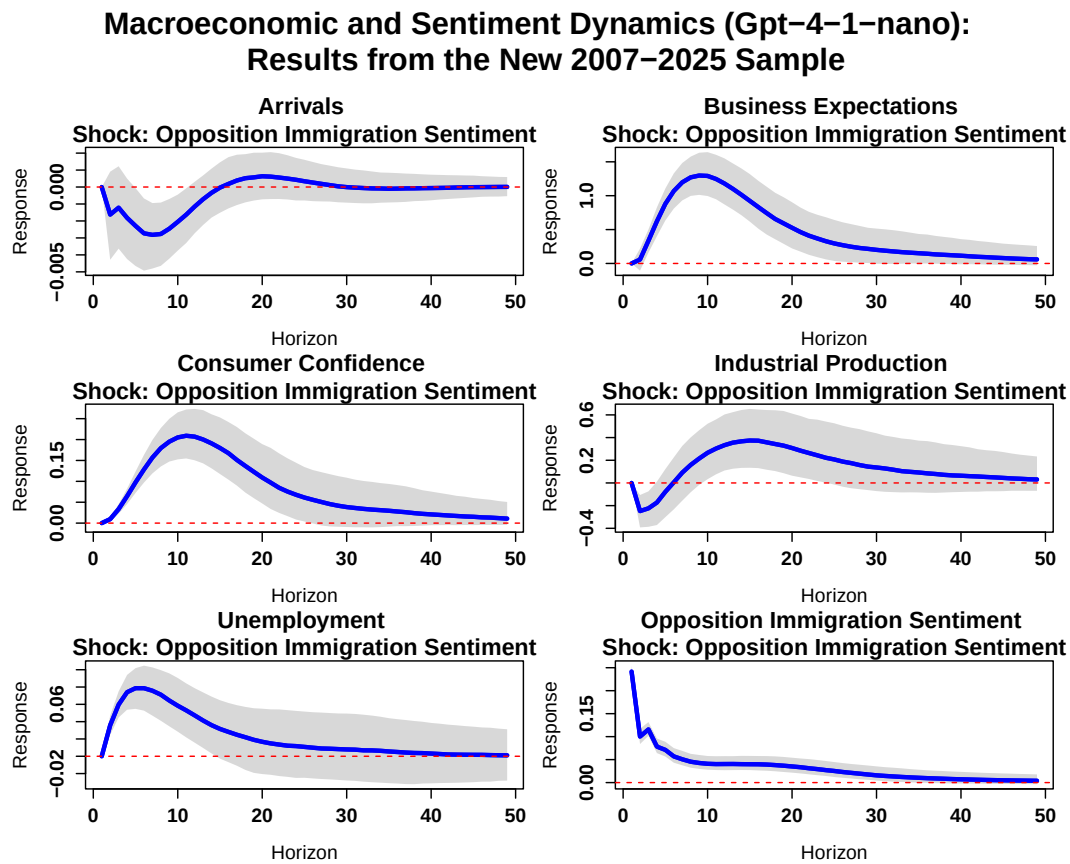


Figure 32: Robustness check, LLM-based sentiment measure, sample 2007–2025. The figure shows the responses of the endogenous variables to an increase in opposition immigration sentiment, constructed using GPT-4.1-nano following Kostikova et al. (2024) (horizons 0–50). Shaded areas denote 68% credible intervals. All impulse responses are obtained from a recursively identified Bayesian VAR model with two lags and Normal–Wishart priors, including a COVID-19 exogenous control dummy for observations between March 2020 and October 2021.

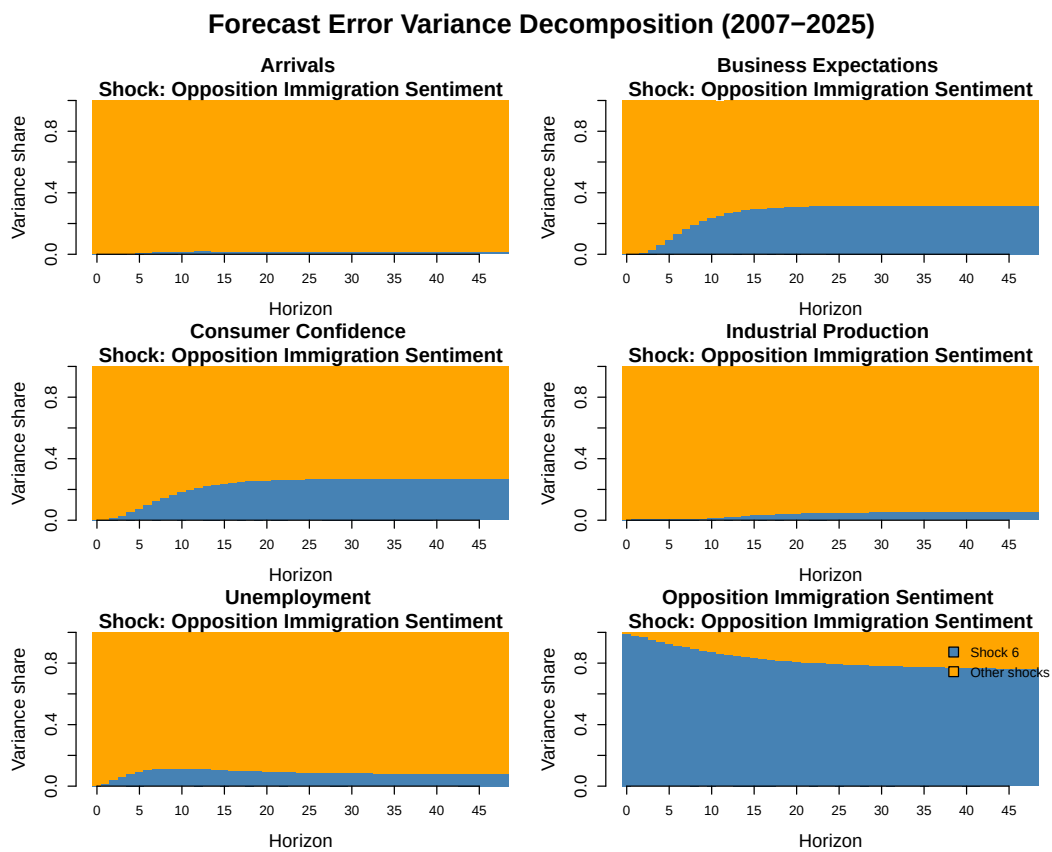


Figure 33: Forecast Error Variance Decomposition (2007–2025) for Opposition Immigration Sentiment Shock

5 Conclusion

This thesis asked whether immigration-related political sentiment in the German Bundestag carries information about later macroeconomic and migration dynamics. To answer this, the analysis combined natural language processing with a Bayesian vector autoregressive (BVAR) framework. The empirical strategy first reproduced the macroeconomic setting of Maffei-Faccioli and Vella (2021) and then extended it by adding a time-varying measure of parliamentary immigration sentiment. The analysis also separated government from opposition sentiment and tested the robustness of the results across sample periods, recursive orderings, and an independent large language model (LLM)-based sentiment measure.

A central methodological contribution is the construction of a dynamic immigration sentiment measure from Bundestag speeches. Instead of relying on a fixed dictionary, the approach combines manually selected seed terms with Word2Vec-based semantic expansion, PCA-guided vocabulary refinement, and rolling-window Doc2Vec representations. The rolling-window design matters because it lets the semantic representation of immigration-related language evolve over time and reduces the risk of leaking future information into the sentiment measure. The resulting index is then split into government, opposition, and aggregate Bundestag sentiment, which lets the analysis check whether political position is linked to different macroeconomic responses.

The baseline BVAR results show that innovations in the dictionary-based sentiment measure are followed by movements in several macroeconomic variables. Sentiment shocks are followed by clear responses in business expectations, consumer confidence, and unemployment, while industrial production responds less strongly. Migration arrivals respond more weakly still, and this response varies across sample periods and political groups. In the reconstructed and extended samples, opposition and aggregate sentiment show especially visible responses in arrivals, unemployment, and expectations-related variables. These findings suggest that immigration-related parliamentary discourse carries economically relevant information beyond its immediate content.

An important finding from the robustness checks is that the empirical relationship depends on how the sentiment measure is built. Three alternative recursive orderings for the 2007–2014 sample give qualitatively similar responses for unemployment,

and to a lesser extent for arrivals, suggesting the main findings are not driven by one specific ordering choice. However, the LLM-based sentiment measure produces a different pattern for several macroeconomic variables. In particular, opposition sentiment is associated with increases in business expectations, consumer confidence, and industrial production in several specifications, while the unemployment response stays more persistent in the extended sample. Arrival responses also differ across sample periods. These differences matter because they show that conclusions about the economic content of political sentiment depend partly on how sentiment is measured.

The comparison between the dictionary-based and LLM-based measures is therefore an important qualification of the results. The dictionary-based measure captures how closely parliamentary language relates to welcoming or restrictive immigration concepts, but it can be distorted by negation, reported speech, and the difference between discussing and endorsing a position. An LLM-based classifier can account for some of these contextual features more directly, though it introduces its own dependence on model choice and classification decisions. The robustness results should not be read as identifying one universally correct sentiment measure. Instead, they show that how political rhetoric is measured is itself a key part of the empirical analysis.

Taken together, the evidence suggests that parliamentary immigration sentiment carries predictive information about later macroeconomic dynamics, but the size and direction of the estimated responses depend on the sentiment measure and sample period. The most persistent result across specifications is the unemployment response, while responses in expectations, industrial production, and arrivals are more sensitive to measurement and sample choice. Second, the responses of the endogenous variables to sentiment shocks built from Doc2Vec and from the LLM, using opposition speeches, show the same pattern over the larger 2007–2025 sample: a positive immigration rhetoric among opposition members is followed by more positive expectations among households and firms, while arrivals and unemployment are more sensitive to the sample period chosen.

One plausible interpretation is that immigration-related rhetoric may partly reflect policymakers' reactions to prevailing economic conditions. For example, labour shortages, changes in migration pressure, or broader economic uncertainty may affect both macroeconomic indicators and how migration is discussed in parliament at the same time. In that case, a positive or welcoming sentiment shock need not be an exoge-

nous policy move; it may instead capture a political reaction to the surrounding economic environment. The empirical design used here cannot fully separate these two mechanisms. This interpretation is offered as one possible explanation of the observed dynamics, not as a causal conclusion.

The thesis also contributes to the literature by showing how political text data can be built into an established macroeconomic framework. Combining parliamentary speech analysis, dynamic semantic representations, and BVAR methods gives a framework for studying political discourse together with economic and migration variables. The government-opposition distinction further shows that parliamentary discourse should not be treated as one uniform block.

Some limitations should be noted. First, the dictionary-based sentiment measure remains vulnerable to contextual features such as negation and reported speech. Second, the reconstructed macroeconomic dataset differs in some respects from the original data used by Maffei-Faccioli and Vella (2021), particularly for consumer confidence, which may explain some of the differences across replication exercises.

These limitations point to directions for future work. Future research could compare several LLMs and classification prompts under one shared validation framework. On the econometric side, richer information sets that include vacancies, wages, investment, consumption, housing prices, and net exports could help separate alternative transmission mechanisms. Nonlinear or asymmetric models could also test whether welcoming and restrictive immigration rhetoric have systematically different economic effects. Extending the analysis to other European parliaments would show whether the link between political immigration discourse and macroeconomic dynamics is specific to Germany or reflects a broader European pattern.

Overall, the findings show that political discourse can be built into quantitative macroeconomic analysis in a systematic way, while also highlighting the methodological challenges of measuring political sentiment from text. The evidence supports the conclusion that immigration-related parliamentary sentiment carries economically relevant information about later macroeconomic and migration dynamics. At the same time, the sensitivity of several impulse responses to the sentiment measure and specification cautions against reading these relationships as stable structural effects. The main contribution of this thesis is therefore not to establish a simple causal link between immigration rhetoric and economic outcomes, but to build and test an empirical

framework through which such relationships can be measured, compared, and studied more rigorously.

Appendix

6 Unit Root and Cointegration Tables

Table 7: KPSS Tests for Level Stationarity

Variable	KPSS	p-value	Decision
Immigration sentiment in government speeches	1.5265	< 0.01	Non-stationary
Immigration sentiment in opposition speeches	1.7362	< 0.01	Non-stationary
Consumer confidence index	1.9268	< 0.01	Non-stationary
Business expectations index	0.1913	> 0.10	Stationary
Industrial production index	1.8047	< 0.01	Non-stationary
Number of registered unemployed	2.9381	< 0.01	Non-stationary
Total net migration	1.7774	< 0.01	Non-stationary

Notes: The macroeconomic variables are based on the dataset employed by Maffei-Faccioli and Vella (Maffei-Faccioli and Vella 2021), extended to the sample period considered in this study. The two political sentiment variables are constructed from parliamentary speeches and distinguish between government and opposition speeches. The KPSS test has level stationarity as the null hypothesis. The truncation lag parameter is 4 for all tests. The null hypothesis is rejected at the 5% significance level when the p-value is below 0.05. Reported p-values of < 0.01 and > 0.10 reflect the bounds returned by the R implementation.

Table 8: Johansen Trace Test for Cointegration

Null Hypothesis	Trace Statistic	10%	5%	1%
$r = 0$	280.88	126.58	131.70	143.09
$r \leq 1$	189.89	97.18	102.14	111.01
$r \leq 2$	117.70	71.86	76.07	84.45
$r \leq 3$	68.57	49.65	53.12	60.16
$r \leq 4$	38.55	32.00	34.91	41.07
$r \leq 5$	17.31	17.85	19.96	24.60
$r \leq 6$	3.66	7.52	9.24	12.97

Notes: The Johansen trace test is estimated with three lags ($K = 3$), a constant restricted to the cointegration space, and no deterministic linear trend. The test is applied to the seven-variable system including migration, business expectations, consumer confidence, industrial production, unemployment, government sentiment, and opposition sentiment. The trace test sequentially tests the null hypothesis of at most r cointegrating relationships. At the 5% significance level, the null hypotheses $r = 0$, $r \leq 1$, $r \leq 2$, $r \leq 3$, and $r \leq 4$ are rejected, whereas $r \leq 5$ cannot be rejected. Therefore, the results indicate a cointegration rank of $r = 5$ among the variables.

Table 9: KPSS Tests for Level Stationarity

Variable	KPSS	p-value	Decision
Immigration sentiment in opposition speeches	3.2246	< 0.01	Non-stationary
Immigration sentiment in government speeches	1.4292	< 0.01	Non-stationary
Consumer confidence index	0.7726	< 0.01	Non-stationary
Business expectations index	1.3439	< 0.01	Non-stationary
Industrial production index	0.7040	0.0132	Non-stationary
Registered unemployment rate	3.3561	< 0.01	Non-stationary
Number of arrivals	0.9843	< 0.01	Non-stationary

Notes: The two political sentiment variables are constructed from parliamentary speeches and distinguish between government and opposition speeches. The KPSS test has level stationarity as the null hypothesis. The truncation lag parameter is 4 for all tests. The null hypothesis is rejected at the 5% significance level when the p-value is below 0.05. Reported p-values of < 0.01 reflect the lower bound returned by the R implementation.

Table 10: Johansen Trace Test for Cointegration

Null Hypothesis	Trace Statistic	10%	5%	1%
$r = 0$	255.62	126.58	131.70	143.09
$r \leq 1$	179.22	97.18	102.14	111.01
$r \leq 2$	121.41	71.86	76.07	84.45
$r \leq 3$	79.56	49.65	53.12	60.16
$r \leq 4$	41.49	32.00	34.91	41.07
$r \leq 5$	18.77	17.85	19.96	24.60
$r \leq 6$	6.16	7.52	9.24	12.97

Notes: The Johansen trace test is estimated with three lags ($K = 3$), a constant restricted to the cointegration space, and no deterministic linear trend. The test sequentially evaluates the null hypothesis of at most r cointegrating relationships. At the 10% significance level, the null hypotheses $r = 0$, $r \leq 1$, $r \leq 2$, $r \leq 3$, $r \leq 4$, and $r \leq 5$ are rejected, whereas $r \leq 6$ cannot be rejected. Therefore, the results indicate a cointegration rank of $r = 6$ among the variables.

7 Additional Impulse Response Figures

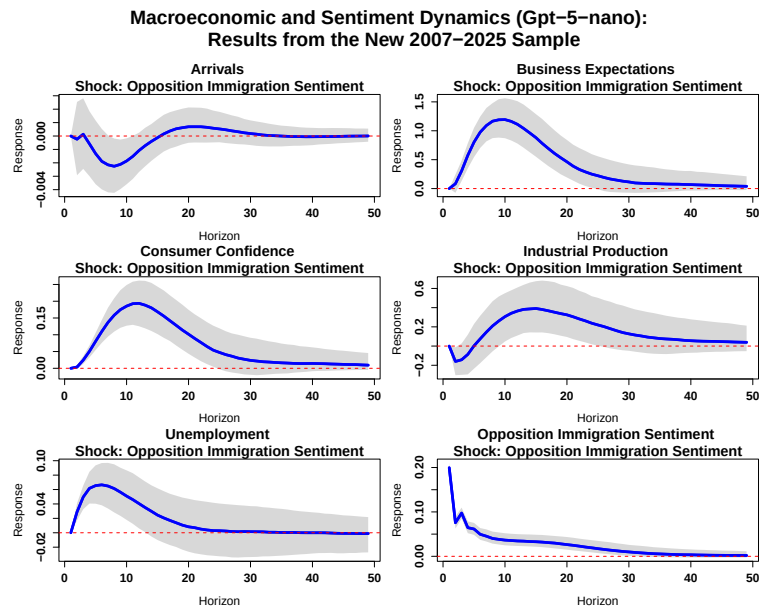


Figure 34: GPT-5-nano, 2007–2025, opposition sentiment shock.

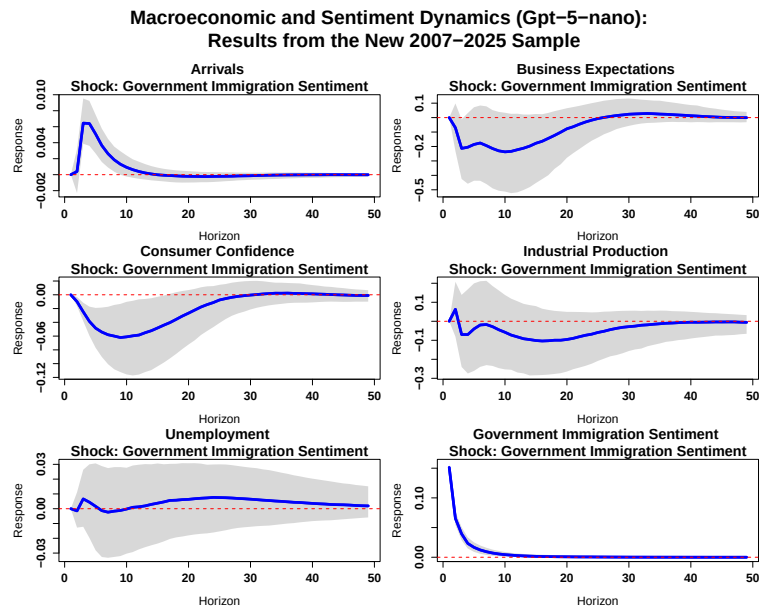


Figure 35: GPT-5-nano, 2007–2025, government sentiment shock.

8 Party-Level Sentiment Statistics

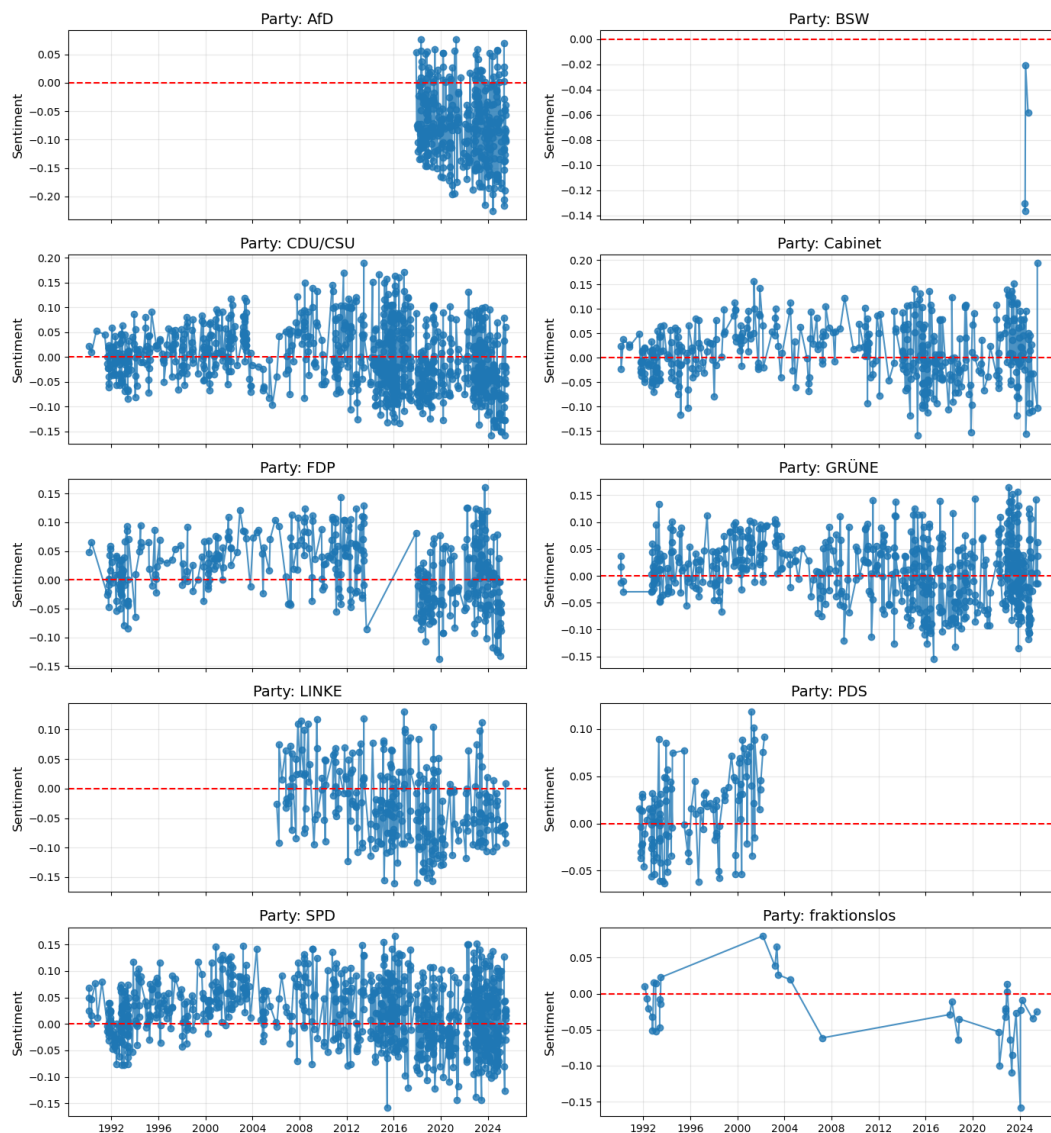


Figure 36: Evolution of Immigration Sentiment by Political Party

Notes: The figure presents the evolution of the estimated immigration sentiment scores for speeches by political party over the sample period. The horizontal red dashed line represents zero sentiment. Positive values indicate relatively pro-immigration sentiment, whereas negative values indicate relatively restrictive sentiment. The figure illustrates differences in the level and variability of immigration-related sentiment across parties and over time.

9 LLM Sentiment Classification Prompt

Sentiment classification via GPT-4.1-nano and GPT-5-nano was performed using the prompt template below, generated programmatically for each speech excerpt. The five

solidarity and five anti-solidarity subtypes, defined in Table 11, are inserted into the template at runtime.

Table 11: Solidarity and anti-solidarity subtypes used in prompt construction.

Subtype	Definition
<i>Solidarity</i>	
Group-based	Migrants are recognized as a legitimate group with equal rights, interests, or claims.
Exchange-based	Support is based on reciprocity: an individual migrant contributes or fulfills obligations and therefore deserves support or rights.
Economic-utility	Migration is presented as beneficial or necessary for the German economy or society (e.g., labor shortages, demographics, growth, taxes, skilled labor).
Compassionate	Migrants/refugees are portrayed as vulnerable or suffering and therefore deserving protection, sympathy, or support.
Empathic	Migrants' identity, dignity, culture, autonomy, or self-determination is explicitly recognized or respected.
<i>Anti-solidarity</i>	
Group-based	Migrants are portrayed as an out-group or as having fewer rights than natives; native interests are prioritized.
Exchange-based	An individual migrant is portrayed as taking more than they contribute or failing to fulfill obligations.
Economic-burden	Migration is presented as economically harmful or costly (e.g., welfare costs, fiscal burden, wage competition, pressure on resources).
Compassionate	Migrants' need for protection or humanitarian support is dismissed or portrayed as undeserving.
Empathic	Migrants' identity, dignity, culture, autonomy, or self-determination is denied or rejected.

Listing 1: Prompt template passed to the LLM for each speech excerpt.

```

Classify the following German parliamentary speech excerpt
    according to its
stance toward migrants, refugees, asylum seekers, or immigrants.

LABELS
-----

SOLIDARITY = supportive, accepting, protective, inclusive, or
    sympathetic.
ANTI-SOLIDARITY = opposing, excluding, rejecting, hostile, or
    unwilling to support.
MIXED = both meaningful solidarity and anti-solidarity.
NONE = no clear stance toward migrants.

RULES
-----
- Migration must be the target. Mentioning migration alone is
    not enough.
- Neutral facts, statistics, questions, or policy descriptions =
    NONE.
- Criticism/support of government or migration policy is not
    automatically
anti/pro-migration; classify only the stance toward migrants.
- Do not classify unrelated groups or historical German
    expellees as migrants.
- If the evidence is unclear, choose NONE.
- If SOLIDARITY or ANTI-SOLIDARITY, choose exactly ONE subtype.
- If MIXED or NONE, subtype = NONE.

SOLIDARITY SUBTYPES
-----
[inserted from Table~\ref{tab:solidarity_subtypes}]

ANTI-SOLIDARITY SUBTYPES
-----
[inserted from Table~\ref{tab:solidarity_subtypes}]

TEXT
-----

```

****<speech excerpt>****

OUTPUT FORMAT

Return EXACTLY two lines and nothing else:

Justification: <one short sentence explaining the decision>

Label: <HIGH-LEVEL>[-<SUBTYPE>]

VALID LABELS

SOLIDARITY-GROUP-BASED

SOLIDARITY-EXCHANGE-BASED

SOLIDARITY-ECONOMIC-UTILITY

SOLIDARITY-COMPASSIONATE

SOLIDARITY-EMPATHIC

ANTI-SOLIDARITY-GROUP-BASED

ANTI-SOLIDARITY-EXCHANGE-BASED

ANTI-SOLIDARITY-ECONOMIC-BURDEN

ANTI-SOLIDARITY-COMPASSIONATE

ANTI-SOLIDARITY-EMPATHIC

MIXED

NONE

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Declaration of originality

I hereby declare that I have written this Masterthesis independently and without using other than the indicated sources and aids and that I have identified the passages taken verbatim or in content from the used sources as such. The work in the same or similar form has not been submitted to any other examination authority.

Linden, 01.09.2026

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